RV College of Engineering

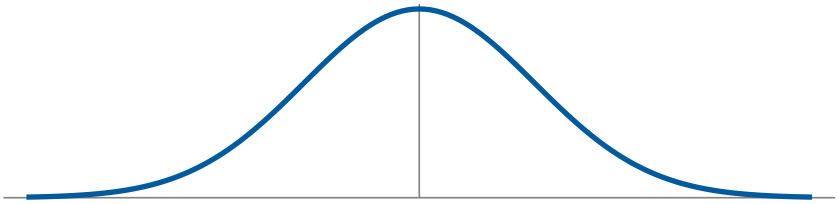
Department of Industrial Engineering and Management

Statistics for Data Analytics

Course Code: IM241AT

Semester IV • Professional Core Course

Unit 1



A Student Compilation for Study and Reference

Semester: IV						
STATISTICS FOR DATA ANALYTICS **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**IM241AT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Data Summary and Presentation: Data types, tabular and graphical displays: Stem and Leaf diagrams, Histograms, Box plots, Radar diagrams. **Concepts of Probability:** Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability, Multiplication and Total probability rules, Independence, Bayes Theorem. Numerical Problems	
Unit-II	**09 Hrs**
Concepts of Random Variables: Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Numerical Problems **Discrete Probability Distributions:** Discrete uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems.	
Unit-III	**09 Hrs**
Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations, Exponential, Applications, Numerical Problems. **Sampling Distribution:** Sampling distribution, Central Limit Theorem, Sampling distribution of means and derived quantities, Numerical Problems.	
Unit-IV	**09 Hrs**
Simple Linear Regression and Correlation: Empirical models, Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Common abuses of regression, Prediction of new observations, Correlation, Numerical Problems. Interpretation of graphical output from software packages such as Minitab	
Unit-V	**09 Hrs**
Statistical Inference for a single sample: Hypothesis testing, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Tests of association, Numerical Problems **Interval Estimation:** Confidence Intervals on mean (variance known and unknown), and variance of a normal population	

Course Outcomes: After completing the course, the students will be able to	
CO1	Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data

References Books	
1.	Engineering Statistics, Douglas C. Montgomery, George C. Runger, Norma FarisHubele, 5th Edition, 2011, John Wiley & Sons, Inc., ISBN-13: 978- 0-470-63147-8
2.	Applied statistics and Probability for Engineers, Douglas C Montgomery, George C Runger, Wiley, 4th Edition, 2007, Asia Student Edition, ISBN: 978-81-265-2315-3.
3.	Statistics for Management, Richard I Levin, David S Rubin, 7th Edition, 1997, Prentice Hall India, ISBN: 9780134762920.
4.	Probability and Statistics for Engineers and Scientists, Walpole, Myers, Myers, Ye, 8th Edition, 2007, Pearson Education Inc., ISBN: 978-81-317-1552-9.
5.	Softwares : Microsoft Excel / Minitab / Matlab / R

6.	Online resources: a) http://172.16.44.44/nnptel.html - choose NOC:Introduction to Data Analytics(Course sponsored by Aricent) b) https://onlinecourses.nptel.ac.in/noc22_mg31/preview -Introduction to probability and Statistics c) https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies d) https://www.khanacademy.org/math/statistics-probability

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40**	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: (Internal Choice)	16
5 & 6	Unit 3: (Internal Choice)	16
7 & 8	Unit 4: (Internal Choice)	16
9 & 10	Unit 5: (Internal Choice)	16
	TOTAL	**100**

A Note on This Compilation

This book is **not** the official RV College of Engineering notes. It is a compilation of student notes together with the handwritten notes provided by **Dr. Ramaa A Nadig**, Department of Industrial Engineering and Management.

All information contained in this book has been drawn from her notes and lectures. The material has been organised, typeset, and arranged to follow the prescribed course syllabus, but it should be treated as a supplementary study aid rather than an official course document.

Every effort has been made to reproduce the source notes faithfully and to preserve their content without addition or deletion. Where the original handwritten working was unclear, the final answers recorded in the notes were treated as authoritative. Readers should always cross-check against the prescribed textbooks and their instructor's guidance.

Compiled from the notes and lectures of Dr. Ramaa A Nadig, Department of Industrial Engineering and Management.

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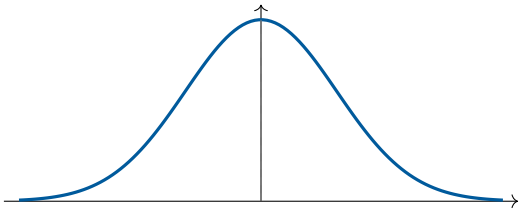
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Unit 1



Chapter 1

Introduction to Engineering Statistics

1.1 What Is Statistics?

At its core, statistics may be defined as *the measurement of uncertainty*. It is the discipline that allows researchers to evaluate conclusions derived from sample data and to quantify how much confidence those conclusions deserve.

Statistics

Statistics is the art and science of collecting, classifying, presenting, interpreting, and analysing numerical data, as well as drawing conclusions about the system from which the data were obtained.

In practice, statistics refers to a scientific approach used to:

- Collect data.
- Interpret and analyse data.
- Assess the reliability of conclusions based on sample data.

A closely related idea is the notion of a single statistic.

Statistic

A **statistic** is a characteristic of a sample. Generally, a statistic is used to estimate the value of a population parameter.

Viewed more broadly, statistics is the collection, analysis, and use of data to understand and predict factors of interest in a scientific way, so as to make decisions, solve problems, and design products or processes. The field of statistics can be broken into two major areas: **descriptive statistics** and **inferential statistics**, both of which are developed in detail later in this unit.

1.2 Why Learn Statistics?

Statistics earns its place in engineering and science for two principal reasons:

- The potential outcome of events can be predicted with the help of statistics in situations where other techniques are not adequate.
- It supports decision making based on what is already known. Familiar examples include the analysis of climate disasters and the forecasting of agricultural production.

1.3 The Role of Statistics in Engineering

Problems of interest are solved by engineers using the scientific method. Statistics provides the formal machinery for the data-driven steps of that method. A typical engineering problem-solving cycle proceeds as follows:

1. Describe the problem.
2. Identify the factors that affect the problem.
3. Propose a model.
4. Collect data from experiments to validate the model.
5. Refine the model based on the data.
6. Develop a solution.
7. Collect data to confirm the solution's effectiveness and efficiency.
8. Conclude and give recommendations.

More specifically, the role of statistics within this cycle includes the following:

- Collection, presentation, analysis, and use of data to make decisions.
- Helping to describe and understand *variability*.
- Making conclusions about a population based on sample observations — the process of *statistical inference*. For example: if sample data exist, are they sufficient to infer something about the population?
- Distinguishing between inherent variability and changes due to deliberate modification of the inputs.

1.4 Applications of Statistics

Statistics is applied across virtually every branch of engineering and the sciences. Representative application areas include:

- **Process improvement** — for example, studying the mean productivity of processes.
- **Product design** — assessing the effect of design parameters.
- **Health care** — for example, evaluating whether a new vaccination programme is going to be effective.
- **Strategic decision making** — decisions based on surveys.
- **Prediction of parameters** — such as financial returns, inflation, and weather.

Numerical data arise in essentially every sphere of human experience. A non-exhaustive catalogue of such data includes: birth and death data; crime data and food-product consumption; agricultural production of different crops; industrial production across heavy, medium, and small-scale industry; earnings through export of various commodities; sports data such as runs scored, wickets taken, goals scored, matches won, and running times; social data such as age at marriage, age of a parent at the birth of a child, and infant-mortality rate; medical data such as the incidence of certain diseases, control and survival rates, the effects and side effects of medicines, and

the relationship between certain diseases and habits; accident data for road, rail, air, and ship, together with fatality rates; counts of vehicles, lengths of roads and rails, and numbers of valid licence holders; numbers of voters and constituencies; astronomical and cosmological data such as planets, the solar system, stars, galaxies, comets, solar and lunar eclipses, and the positions of certain stars and planets; chemical compositions and the study of electrons and neutrons in metallurgical work; genetic studies of human behaviour; biometric studies; the percentage of defective items and the number of items produced; data on consumers of certain products, their capacity to spend, their income levels, and profession; weather data; and placement data.

1.5 A Brief History of Statistics

1.5.1 Western Developments

The development of statistics as a discipline can be traced through a sequence of milestones:

- The word *statistics* was coined by the German scholar Gottfried Achenwall around the middle of the eighteenth century. It is derived from the word *state*.
- It was used for the first time in Britain by John Sinclair (1791–1799).
- The term appeared in the *Encyclopaedia Britannica* in 1797.
- The earliest work on statistical methodology was related to the normal distribution (De Moivre, 1733; the paper was later discovered by Karl Pearson in 1924).
- The French mathematician Laplace, in his manuscript of 1783, stated that the normal curve can be used for describing the probability distribution of errors.
- In 1809, Gauss developed the normal distribution in a study of the theory of the motions of heavenly bodies; hence the alternative name *Gaussian distribution*.
- The Belgian mathematician Quetelet (1796–1874) applied statistics to such quantities as the heights of a large number of people.
- Gregor Mendel performed various genetic experiments, mostly on peas, and concluded that the laws of heredity are statistical in nature.
- Francis Galton, in *Hereditary Genius* (1869) and *Natural Inheritance* (1889), argued that hereditary laws are statistical in nature, and so introduced the concepts of correlation and regression.
- Karl Pearson introduced the concept of statistics in a systematic way. In 1901 he started the famous journal *Biometrika*, one of the top journals for the study of statistics. He was the first to discuss the chi-square test statistic, used for testing data for the normal distribution or a gamma distribution — the test of goodness of fit.
- G. U. Yule, in 1910, gave the distinction between a population and a random sample; later, R. A. Fisher also gave a clear-cut distinction between population and sample.
- The definition of random sampling was given by John Venn and C. S. Peirce.
- The first table of random digits was printed by Tippett in 1927.
- Gosset ("Student") contributed the t-distribution.

- R. A. Fisher is regarded as the father of modern statistics. He developed the theory and foundations of statistical inference, maximum-likelihood estimation, unbiased estimation, and the analysis of variance.
- Neyman and E. S. Pearson started developing the testing of hypotheses.
- Fisher and Neyman are credited with giving confidence intervals; Neyman further developed stratified random sampling.
- In Russia, around 1939, Kolmogorov developed a rigorous theory of probability and stochastic processes.

1.5.2 Indian Developments

- Kautilya's *Arthashastra*, written in 321 BC, gives a detailed description of how to collect data on various aspects of the state.
- Modern statistical methodology in India was initiated by P. C. Mahalanobis, who studied in England and came back to India to use statistical tools regularly. With his efforts, the Indian Statistical Institute was started in 1931.
- R. C. Bose contributed to the subject of the design of experiments.
- S. N. Roy and C. R. Rao contributed greatly to multivariate analysis, the design of experiments, and linear models.
- With the efforts of Mahalanobis, the Central Statistical Organisation was established in 1951 by the Government of India to collect data on various aspects of the state. In 1950, the National Sample Survey Organisation was established to collect data on various socio-economic aspects of life.

1.6 Collecting Engineering Data

Engineering data are collected in three principal ways:

- A retrospective study using historical data.
- An observational study.
- A designed experiment.

1.6.1 Retrospective Study

A retrospective study would use either all or a sample of the historical process data archived over some period of time. This study may involve a large amount of data, but that data may contain relatively little useful information about the problem. Furthermore, some of the relevant data may be missing, and there may be transcription or recording errors resulting in outliers. As a result of these types of issues, statistical analysis of historical data sometimes identifies interesting phenomena, but solid and reliable explanations of these phenomena are often difficult to obtain.

1.6.2 Observational Study

In an observational study, the engineer observes the process or population, disturbing it as little as possible, and records the quantities of interest.

1.6.3 Designed Experiment

In a designed experiment, the engineer makes deliberate or purposeful changes in the controllable variables of the system or process, observes the resulting system output data, and then makes an inference or decision about which variables are responsible for the observed changes in output performance.

1.7 Models

Mathematical Model

A **mathematical model** can be defined as a description of a system using mathematical concepts and languages, in order to facilitate proper explanation of a system, to study the effects of different components, and to make predictions on patterns of behaviour.

Three useful categories of model are distinguished below.

Mechanistic model. A mechanistic model is built from underlying knowledge of the basic physical mechanism that relates the variables of interest. It is a description of a complex system based on a description of its components and their interconnections; such models mostly have a tangible, physical aspect.

Empirical model. An empirical model uses our engineering and scientific knowledge of the phenomenon, but it is not directly developed from theoretical or first principles of the underlying mechanism. In general, one example of an empirical model is regression. To estimate the parameters, the method of least squares is used, which traces its origin to work by Karl Gauss.

Probability model. Probability models help quantify the risks involved in statistical inference — that is, the risks involved in decisions made every day.

Example 1.1 (Two ways to model the tides)**.** Consider the prediction of tides. There are two contrasting modelling philosophies:

- **Empirical** — based on past data, without explicitly knowing the mechanism.
- **Mechanistic** — based on the physical interaction of the Earth, Sun, and Moon.

1.8 Raw and Processed Data

Data may be classified by the degree of processing applied to them.

- **Raw data** — data as collected, without further processing.
- **Processed data** — data processed after collection.

The distinction is best appreciated through paired examples.

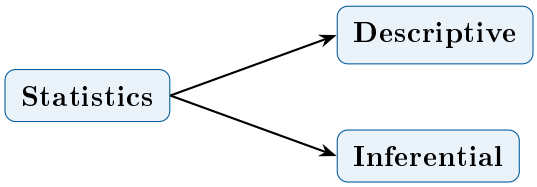
Table 1.1: Raw data versus processed data

Raw	**Processed**
Height of all students in a particular class.	Average height of each class of the IV semester.
Online measurement of the weight of each pack of a food item.	Report containing the average weight of a pack for a food item per day.
Weight of a casting part measured at different manufacturing looms in different units of measurement.	Weight of components converted to a single unit for comparison purposes.
Downtime in days of a machine in a factory or shop floor.	Inflation rates for the past 10 years; yearly sales figures of automobile manufacturers in India.
Length of phone calls for 200 persons calling through a particular exchange.	—

Chapter 2

Descriptive and Inferential Statistics

The field of statistics divides naturally into two complementary branches. Descriptive statistics is concerned with *summarising* data that are in hand, whereas inferential statistics is concerned with *generalising* from a sample to a larger population. Probability is the conceptual bridge that links the two.



2.1 Descriptive Statistics

Descriptive Statistics

Descriptive statistics is the branch of statistics which characterises and summarises the most prominent features of a given set of data.

Some of the commonly used descriptive measures include means, medians, standard deviations, percentiles, graphs, tables, and charts. Descriptive statistics can be used to describe the elements of a population as a whole, or they can be used to describe data that represent just a sample of elements from the entire population.

2.2 Inferential Statistics

Inferential Statistics

Inferential statistics is the branch of statistics that deals with drawing conclusions about a population based on information obtained from a sample drawn from that population.

2.3 Probability: The Bridge Between the Two Branches

Probability is the bridge between descriptive statistics and inferential statistics. It is also a link between a population and a sample. When certain properties of a population are known or can be assumed, the probability principles involved allow one to deduce the expected properties of a sample. Conversely, probability together with sample data allows one to evaluate the amount of uncertainty when making conclusions about a population. Hence probability is the heart of both the deductive and inferential processes involved in statistical analysis.

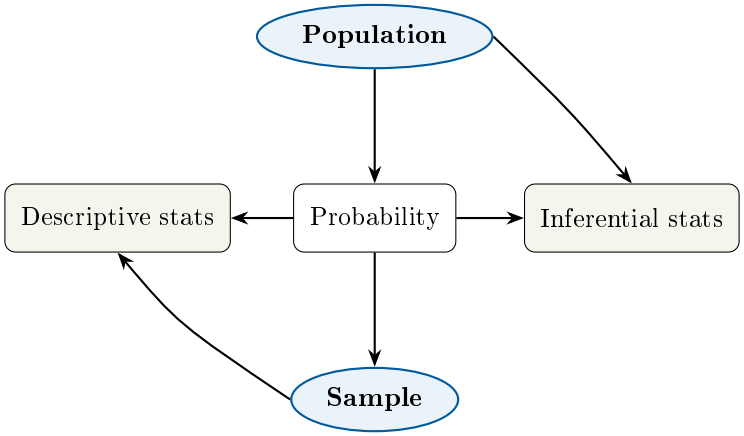


Figure 2.1: Probability links a population to its samples and connects descriptive and inferential statistics. Knowledge of the population, processed through probability, yields expected properties of a sample; sample data, processed through probability, support inferences about the population.

Chapter 3

Types of Data

Before any data can be summarised or analysed, they must be classified by type, because the type of data determines which summaries, visualisations, and operations are meaningful. Data are first divided into **numerical** and **categorical** types, each of which is further subdivided.

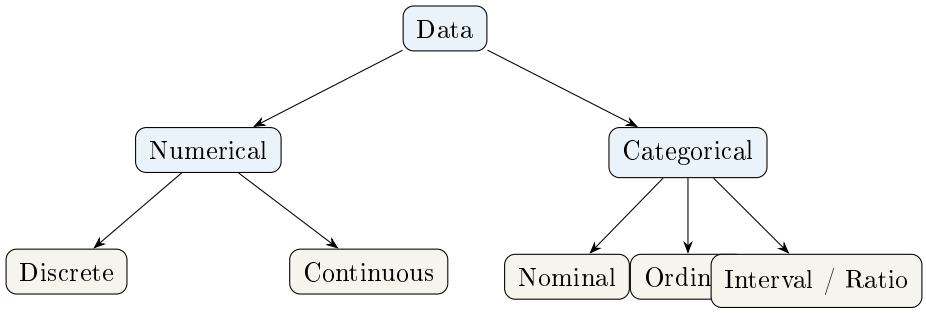


Figure 3.1: Classification of data into numerical and categorical types and their subtypes.

The four measurement scales — nominal, ordinal, interval, and ratio — form a hierarchy of increasing mathematical structure. Nominal and ordinal scales describe categorical data; interval and ratio scales describe numerical data.

3.1 Categorical Data

Categorical data represent characteristics. They can therefore represent things such as a person's gender or the language they speak. Categorical data can also take on numerical values — for example, 1 for female and 0 for male — but it is essential to note that those numbers do not have any mathematical meaning.

3.1.1 Nominal Data

Nominal Data

Nominal data represent discrete units and are used to label variables that have no quantitative value. Nominal data have no order.

Example 3.1 (Nominal variables)**.** • **Gender:** Male, Female. (A variable with only two categories, such as gender, is called *dichotomous*.)

• **Language spoken:** French, English, German, Spanish.

3.1.2 Ordinal Data

Ordinal Data

Ordinal values represent discrete and ordered units. Ordinal data are therefore the same as nominal data except that the ordering matters.

Example 3.2 (Educational background)**.** The categories

1. Elementary 2. High school 3. Undergraduate (UG) 4. Postgraduate (PG)

are ordinal: the order is important, but the difference between the values is not really known.

3.2 Numerical Data

3.2.1 Discrete Data

Discrete Data

Discrete numerical data cannot be measured but can be counted.

Example 3.3. The number of heads in 100 coin flips is discrete data.

3.2.2 Continuous Data

Continuous Data

Continuous data represent measurements; therefore their values cannot be counted but can be measured.

Example 3.4. The height of a person is continuous data.

3.2.3 Interval Data

Interval Data

Interval data represent ordered units that have the same difference between successive values.

Example 3.5. Temperature measured on a scale such as $-10,\ -5,\ 0,\ +5,\ +10$ is interval data.

The problem with interval values is that they do not have a true zero. As a consequence, with interval data we can add and subtract, but we cannot multiply, divide, or calculate ratios.

3.2.4 Ratio Data

Ratio Data

Ratio data are also ordered units that have the same difference between successive values. Ratio data are the same as interval values, with the crucial difference that they *do* have an absolute zero.

Example 3.6. Length in inches, such as $0,\ 5,\ 10,\ 15$, is ratio data. Because length has an absolute zero, ratios between lengths are meaningful.

3.3 Data Collection and Visualisation by Type

The admissible summaries and visualisations depend on the data type.

- **Nominal data** — summarised by frequency, proportion, and percentage; visualised with the pie chart and the bar chart.
- **Ordinal data** — visualised with the pie chart and bar chart, and additionally summarised using percentiles, median, mode, and the interquartile range.

Chapter 4

Graphical Presentation of Data

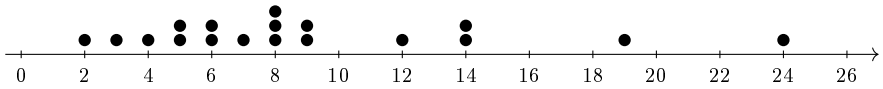
Graphical methods reveal features of a data set — its centre, spread, extremes, gaps, and possible associations — that are difficult to perceive from a raw list of numbers. This chapter develops the dot plot, the scatter plot, the frequency distribution and histogram, the box plot, the stem-and-leaf plot, and the radar chart, and illustrates each with worked problems.

4.1 The Dot Plot (Univariate)

When only one variable is considered and plotted, dot plots are used. They are a simple, attractive display that reveal the extremes, the centering, and the voids or gaps in the data.
Example 4.1 (Constructing a dot plot)**.** Suppose the data points are

$$12,\ 14,\ 9,\ 8,\ 7,\ 8,\ 4,\ 5,\ 14,\ 19,\ 2,\ 3,\ 6,\ 6,\ 5,\ 8,\ 9,\ 24.$$

The corresponding dot plot is shown below.



Dot plots are used when the number of units is relatively small — say $n < 30$ — or where there are relatively few distinct values represented in the data set. In such cases, dot plots are effective displays.

4.2 The Scatter Plot (Bivariate)

When we wish to jointly display results for two variables, the bivariate equivalent of the dot plot is called the **scatter plot**. Similarly, a three-dimensional scatter plot is used with three variables.

4.3 Frequency Distribution and Histogram

A frequency distribution is a more useful summary of data. To construct a frequency distribution, we must divide the range of the data into intervals, which are usually called **class intervals**. If possible, the class intervals should be of equal width, to enhance the visualisation.

The number of class intervals used depends on the number of observations and the amount of scatter or dispersion in the data. Generally, 5 to 20 class intervals are found satisfactory in most cases, and the number of class intervals should increase with n. Choosing a number of class intervals approximately equal to the square root of the number of observations often works well.

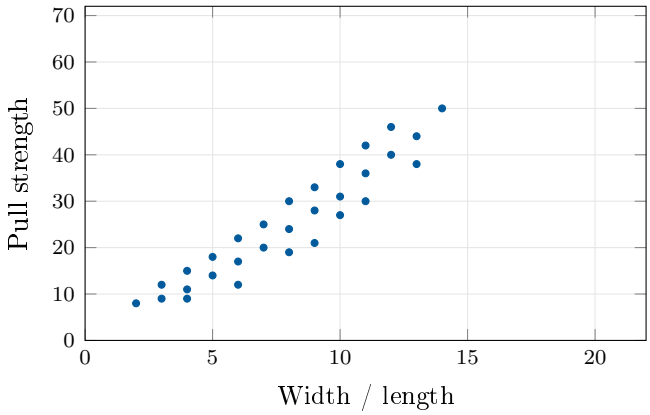


Figure 4.1: A scatter plot. Scatter plots show the region of the plane where data points fall, as well as the data density associated with that region. They also suggest possible association between the variables. Unlike dot plots, scatter plots can be used for large data sets as well.

Choosing the number of classes

Let n be the number of observations and let k denote the number of classes. A practical rule is

$$k \approx \sqrt{n} \quad (n \leq 100), \qquad k \approx 1 + 3.322 \log_{10} n \quad (n > 100),$$

where the second expression is **Sturges' formula**. More generally, $6 \leq k \leq 15$ classes is recommended.

Remark 4.1 (Requirements on histogram classes)**.** For a histogram, the classes should be *all-inclusive* and *mutually exclusive*: every observation falls in exactly one class.

A histogram of cumulative frequency or cumulative relative frequency is called an **ogive curve**.

Worked Problem 1: Shelf Life of Photographic Film

Example 4.2 (Histogram and summary statistics)**.** The shelf life of a high-speed photographic film is being investigated by the manufacturer. The following data (life in days) are available:

126	131	116	125	134	120	125	150	130	149
129	132	128	126	127	122	111	148	120	117
134	136	130	134	120	129	147	126	117	143
141	145	162	129	127	133	129	140	131	133

Here the minimum is 111 and the maximum is 162, so the data span roughly 110 to 170. Construct the histogram and comment on the properties of the data.

The frequency table, using class intervals of width 10, is as follows.

The summary statistics for these data are:

$$\text{Mean } \bar{x} = \frac{\sum x_i}{n} = 131.30, \qquad \text{Median} = 129.5, \qquad \text{Mode} = 129,$$

$$s^2 = 113.85, \qquad s = 10.67.$$

The sample variance is computed from

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1}.$$

Table 4.1: Frequency distribution for the shelf-life data

Class interval	Frequency	Relative frequency	Cumulative rel. freq.
$110 \le x < 120$	4	0.100	0.100
$120 \le x < 130$	16	0.400	0.500
$130 \le x < 140$	11	0.275	0.775
$140 \le x < 150$	8	0.200	0.975
$150 \le x < 160$	0	0.000	0.975
$160 \le x < 170$	1	0.025	1.000
Total	**40**	**1.000**	

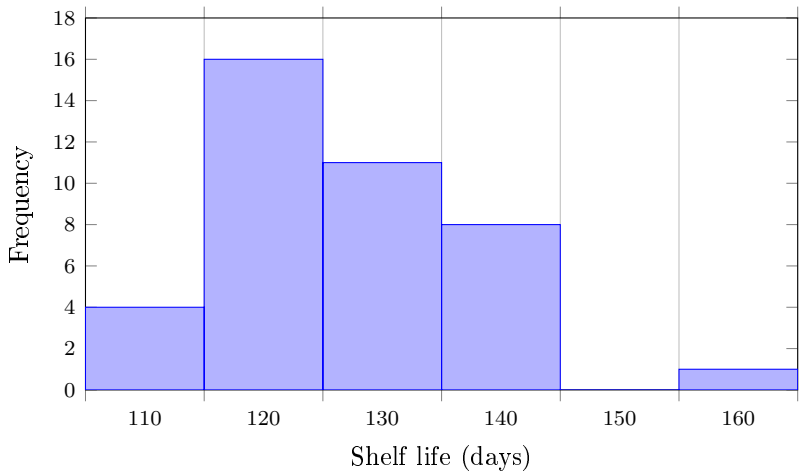


Figure 4.2: Histogram of the shelf-life data. The distribution is unimodal, slightly right-skewed, centred near 130 days, with a single isolated large value near 162.

Worked Problem 2: Number of Defects in Parts

Example 4.3 (Histogram, box plot, and measures of centre)**.** 62 parts were analysed for the number of defects, with the following distribution:

Number of defects i	2	3	4	5	6	7	10
Number of parts with i defects	30	15	9	4	2	1	1

(i) Plot a histogram for the number of defects using 5 classes. (ii) Construct a box plot for the number of defects in a part. (iii) Compute the mean, median, and mode for the number of defects in a part.

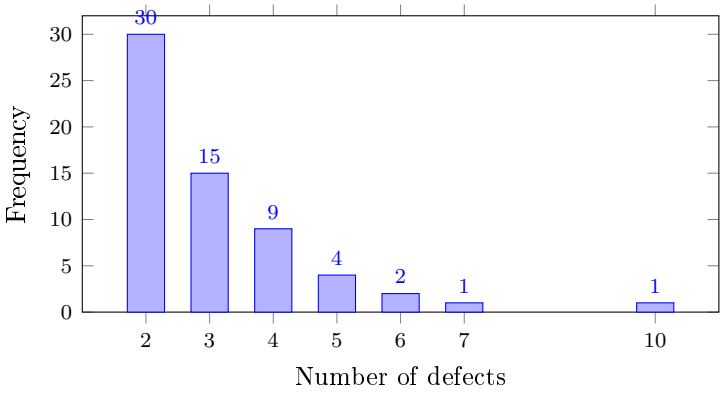


Figure 4.3: Histogram for the number of defects per part.

Histogram.

Measures of centre. The mean is

$$\bar{x} = \frac{2(30) + 3(15) + 4(9) + 5(4) + 6(2) + 7(1) + 10(1)}{62} = \frac{190}{62} = 3.06.$$

The mode is 2, the most frequently occurring value.

Quartiles and the box plot. The cumulative frequencies are

$$30,\ 45,\ 54,\ 58,\ 60,\ 61,\ 62.$$

Using $\dfrac{n+1}{4} = \dfrac{63}{4} = 15.75$, the quartile positions are

$$Q_1 = 15.75^{\text{th}}, \quad Q_2 = 31.5^{\text{th}}, \quad Q_3 = 47.25^{\text{th}} \text{ position},$$

giving

$$Q_1 = 2, \quad Q_2 = 3, \quad Q_3 = 4, \qquad \text{IQR} = 4 - 2 = 2.$$

The fences are

$$\text{lower fence} = 2 - 1.5(2) = -1, \qquad \text{upper fence} = 4 + 1.5(2) = 7.$$

Hence values beyond 7 (here, the value 10) are flagged as outliers.

4.4 Percentiles, Quartiles, and the Box Plot

The box plot rests on the notions of percentiles and quartiles, which locate positions within an ordered data set.

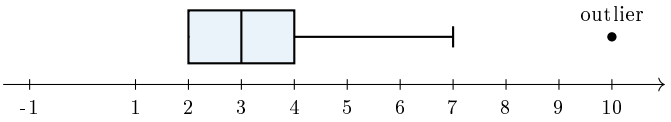


Figure 4.4: Box plot for the number of defects. The box spans $Q_1 = 2$ to $Q_3 = 4$ with the median at 3; the value 10 lies beyond the upper fence and is an outlier.

Quartiles and interquartile range

The **first**, **second**, and **third quartiles** Q_1, Q_2, Q_3 divide ordered data into four equal parts. The second quartile Q_2 is the median. The **interquartile range** is

$$\mathrm{IQR} = Q_3 - Q_1.$$

To locate quartiles by rank, order the data and use positions such as $(n+1)/4$, $(n+1)/2$, and $3(n+1)/4$. For instance, with 100 numbers, the median is the average of the two observations with ranks 50 and 51; the tenth percentile is the observation with rank $(10+11)/2 = 10.5$; the first quartile lies at rank $(25+26)/2 = 25.5$; and the third quartile lies at rank 75.5.

The box plot

The **box plot** displays the three quartiles together with the minimum and maximum of the data on a rectangular box, aligned either horizontally or vertically. The box encloses the interquartile range, with the left line at the first quartile Q_1 and the right line at the third quartile Q_3. A line is drawn through the box at the second quartile (the 50th percentile, or median) $Q_2 = \tilde{x}$. Lines called **whiskers** extend from the ends of the box to the extreme values. In large data sets the whiskers may extend only up to the 10th and 90th percentiles, or the 5th and 95th percentiles.

The box plot is especially useful in comparing two or more samples. Outliers are conventionally identified using fences placed at a distance of 1.5 IQR beyond each quartile.

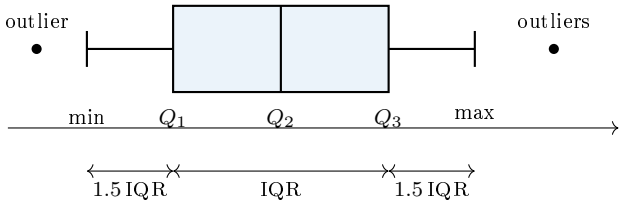


Figure 4.5: Anatomy of a box plot. The box spans the interquartile range from Q_1 to Q_3, the central line marks the median Q_2, and whiskers reach the most extreme non-outlier values. Points beyond 1.5 IQR of the quartiles are outliers.

Worked Problem 3: Furnace Temperatures

Example 4.4 (Mean, variance, median, and box plot)**.** The nine measurements that follow are furnace temperatures recorded on successive batches in a semiconductor manufacturing process (units are in °F):

$$953,\ 950,\ 948,\ 955,\ 951,\ 949,\ 957,\ 954,\ 955.$$

(a) Calculate the sample mean, variance, and standard deviation. (b) Find the median. How much could the largest temperature measurement increase without changing the median value?

(c) Construct a box plot of the data.

(a) Mean, variance, standard deviation.

$$\bar{x} = \frac{953 + 950 + 948 + 955 + 951 + 949 + 957 + 954 + 955}{9} = 952.444.$$

$$s^2 = \frac{(953 - 952.44)^2 + \cdots + (955 - 952.44)^2}{9 - 1} = 9.53, \qquad s = \sqrt{9.53} = 3.0867.$$

(b) Median. Sorting in ascending order:

$$948,\ 949,\ 950,\ 951,\ 953,\ 954,\ 955,\ 955,\ 957.$$

The number of measurements is odd, so the median is the middle value of the sorted data set:

$$M = Q_2 = 953.$$

There is no limit to which the highest temperature measurement can increase without changing the median value, because the middle value is not affected by the largest value in the data set.

Quartiles.

$$Q_1 = \frac{n+1}{4} = \frac{10}{4} = 2.5^{\text{th}} \text{ value} \Rightarrow Q_1 = 949.5,$$

$$Q_2 = 5^{\text{th}} \text{ value} = 953, \qquad Q_3 = 7.5^{\text{th}} \text{ position} = 955.$$

$$\text{Range} = 957 - 948 = 9, \qquad \text{IQR} = Q_3 - Q_1 = 955 - 949.5 = 5.5.$$

$$\text{Lower fence} = 949.5 - 1.5(5.5) = 941.25, \qquad \text{Upper fence} = 955 + 1.5(5.5) = 963.25.$$

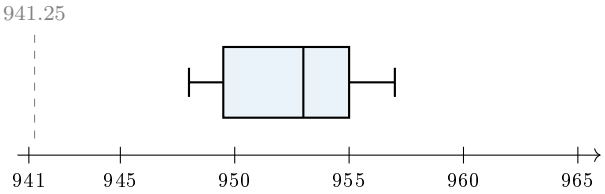


Figure 4.6: Box plot of the furnace-temperature data. No points fall beyond the fences, so there are no outliers.

(c) Box plot.

Worked Problem 4: A Ten-Point Data Set

Example 4.5. For the data

$$38,\ 39,\ 41,\ 45,\ 46,\ 47,\ 48,\ 48,\ 51,\ 52 \qquad (n = 10),$$

the quartiles are located as follows:

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} = \left(\frac{11}{4}\right)^{\text{th}} = 2.75^{\text{th}} \text{ value} = 40.5,$$

$$Q_2 = \left(\frac{n+1}{2}\right)^{\text{th}} = 5.5^{\text{th}} \text{ value} = 46.5,$$

$$Q_3 = \frac{3(n+1)^{\text{th}}}{4} = \frac{11 \times 3}{4} = 8.25^{\text{th}} \text{ value} = 48.75.$$

$$\text{IQR} = Q_3 - Q_1 = 48.75 - 40.5 = 8.25.$$

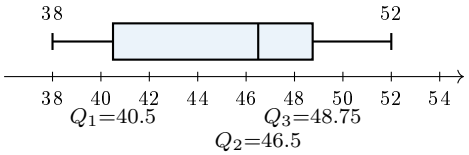


Figure 4.7: Box plot for the ten-point data set of Problem 4.

4.5 The Stem-and-Leaf Plot

Stem-and-leaf plot

Suppose that the data are represented by $x_1, x_2, \ldots, x_n$ and that each number x_i consists of at least two digits. To construct a **stem-and-leaf plot**, we divide each number x_i into two parts: a **stem**, consisting of one or more of the leading digits, and a **leaf**, consisting of the remaining digit. For example, if the number is 76, then we could divide it into stem 7 and leaf 6.

In general, choose few stems compared to the number of observations. It is best to choose between 5 and 20 stems. Once a set of stems has been chosen, they are listed along the left-hand margin of the display, and beside each stem all leaves corresponding to the observed data values are listed in the order in which they are encountered in the data set.

In order to assist in finding percentiles, we order the leaves by magnitude, producing an **ordered stem-and-leaf plot**. The first and third quartiles are occasionally denoted by the symbols Q_1 and Q_3 respectively, and the interquartile range is $\text{IQR} = Q_3 - Q_1$.

Worked Problem 5: Defects in Twenty Parts

Example 4.6. The number of defects when 20 parts were analysed were recorded as

35, 18, 20, 29, 21, 8, 16, 14, 13, 25, 49, 21, 38, 18, 92, 19, 19, 15, 22, 29.

(a) Draw the box plot. (b) Draw the stem-and-leaf diagram. What inferences can be drawn from these plots?

Ordered stem-and-leaf plot.

Stem	Leaf	Frequency
0	8	1
1	3 4 5 6 8 8 9 9	8
2	0 1 1 2 5 9 9	7
3	5 8	2
4	9	1
5		0
6		0
7		0
8		0
9	2	1

Inference: there are large gaps in the data, with minimum 8 and maximum 92.

Box plot. Arranging in ascending order:

$$8,\ 13,\ 14,\ 15,\ 16,\ 18,\ 18,\ 19,\ 19,\ 20,\ 21,\ 21,\ 22,\ 25,\ 29,\ 29,\ 35,\ 38,\ 49,\ 92.$$

$$Q_2 = \left(\frac{n+1}{2}\right)^{\text{th}} = 10.5^{\text{th}} \text{ position} = 20.5,$$

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} = 5.25^{\text{th}} \text{ position} = 16.5,$$

$$Q_3 = \frac{3(n+1)}{4}^{\text{th}} = 15.75^{\text{th}} \text{ position} = 29.$$

$$\text{IQR} = 29 - 16.5 = 12.5.$$

$$\text{Lower fence} = 16.5 - 12.5(1.5) = -2.25, \qquad \text{Upper fence} = 29 + 12.5(1.5) = 47.75.$$

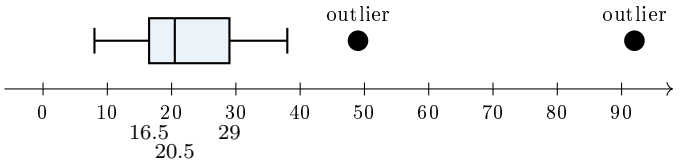


Figure 4.8: Box plot for Problem 5. **Interpretation:** the data set is unsymmetric; the middle 50% of the data lie between 16.5 and 29. The value 49 is a suspected outlier, and 92 is an outlier.

Worked Problem 6: Twenty Weight Measurements

Example 4.7. 20 weight measurements in a process were recorded as

$$2,\ 8,\ 11,\ 10,\ 8,\ 4,\ 2,\ 2,\ 9,\ 8,\ 4,\ 11,\ 12,\ 2,\ 4,\ 4,\ 20,\ 1,\ 10,\ 15.$$

(i) Draw a box plot for the data, showing the fences, quartiles, max, and min clearly. (ii) What are the range and the IQR? (iii) Are there any outliers?

Sorted data.

$$1,\ 2,\ 2,\ 2,\ 2,\ 4,\ 4,\ 4,\ 4,\ 8,\ 8,\ 8,\ 9,\ 10,\ 10,\ 11,\ 11,\ 12,\ 15,\ 20.$$

$$Q_1 = \frac{21}{4} = 5.25^{\text{th}} \text{ position} = 2.5, \quad Q_2 = \frac{21}{2} = 10.5^{\text{th}} = 8, \quad Q_3 = \frac{3(21)}{4} = 15.75^{\text{th}} = 10.75.$$

$$\text{Range} = 20 - 1 = 19, \qquad \text{IQR} = 10.75 - 2.5 = 8.25.$$

$$\text{Lower fence} = 2.5 - 1.5(8.25) = -9.875, \qquad \text{Upper fence} = 10.75 + 1.5(8.25) = 23.125.$$

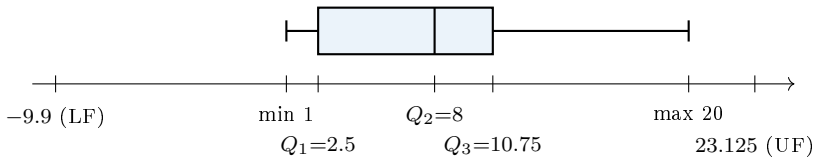


Figure 4.9: Box plot for Problem 6. Since both the minimum and maximum lie within the fences, there are **no outliers**.

Worked Problem 7: Motor-Fuel Octane Ratings

Example 4.8. The data on the motor-fuel octane ratings of several blends of gasoline are given in the table below. Construct a stem-and-leaf diagram and highlight the important feature that can be noticed from it.

93	90.2	91	92.4	90.5	88.7	89.8	89.5
89.8	84.6	91	87.3	92.1	94.8	90.2	89.8
88.8	96.7	88.7	91.3	85.9	85.6	84.7	89.3
89.0	89.4	92.6	87.7	92.3	92.3	86.5	
89.5	87.3	88.7	94.1	87.2	85.4	90.7	

Here the minimum is 84 and the maximum is 96.

Ordered stem-and-leaf plot (stems are integer ranges, leaves are tenths).

Stem	Ordered leaf	Cum. freq.
84	6 7	2
85	4 6 9	5
86	5	6
87	2 3 3 7	10
88	7 7 7 8	14
89	0 3 4 5 5 7 8 8	(8)
90	2 2 5 7	16
91	0 0 3	12
92	1 3 3 4 6	9
93	0	4
94	1 8	3
95		
96	7	1

The important feature is that the bulk of the octane ratings concentrate in the high-80s to low-90s, with the modal stem at 89.

Worked Problem 8: Furnace Temperatures (Revisited)

This problem repeats the nine furnace-temperature measurements of Problem 3, with identical data

$$953,\ 950,\ 948,\ 955,\ 951,\ 949,\ 957,\ 954,\ 955,$$

and asks again for (a) the sample mean, variance, and standard deviation; (b) the median and the amount by which the largest measurement may increase without changing the median; and (c) a box plot. The full solution is given in Problem 3; the principal results are repeated for convenience:

$$\bar{x} = 952.444, \quad s^2 = 9.53, \quad s = 3.0867, \quad M = Q_2 = 953,$$

$$Q_1 = 949.5,\ Q_3 = 955,\ \text{Range} = 9,\ \text{IQR} = 5.5,$$

$$\text{Lower fence} = 941.25, \quad \text{Upper fence} = 963.25.$$

As before, the median is unaffected by increases in the largest value, so there is no upper limit on how much the maximum may grow without changing the median.

Worked Problem 9: History Test Scores

Example 4.9. Observe the stem-and-leaf diagram of history test scores (July 2017) and answer the following.

Stem	Leaf
6	1 1 4 6 7 8
7	2 3 5 7 9
8	1 3 5 6 6 7 7 8 9
9	0 0 3 4 6 8 9 9
10	0 0

(i) **What is the best score?** 100.

(ii) **How many students took the test?** 30.

(iii) **How many students scored 90?** 2.

(iv) **What is the lowest score?** 61.

(v) **Find the range for the scores.** $100 - 61 = 39$.

Worked Problem 10: Process Times in a Chemical Industry

Example 4.10. Process times in a chemical industry are recorded in hours as

4.2, 4.7, 4.7, 5.0, 3.8, 3.6, 3.0, 5.1, 3.1, 3.8, 4.8, 4.0, 5.2, 4.3, 2.8, 2.0, 2.8, 3.3, 4.8, 5.0.

(i) Calculate the sample mean and standard deviation. (ii) Construct a box plot of the data. (iii) Construct a stem-and-leaf plot and comment on the data.

Mean and standard deviation.

$$\bar{x} = \frac{80}{20} = 4.0, \qquad s^2 = \frac{16.46}{19} = 0.866, \qquad s = 0.9307.$$

Ordered stem-and-leaf plot (stem = units, leaf = tenths).

Stem	Ordered leaf	Freq.
2	0 8 8	3
3	0 1 3 6 8 8	6
4	0 2 3 7 7 8 8	(7)
5	0 0 1 2	4

Quartiles.

$$Q_1 = \left(\frac{1}{4}\right) \times 21 = 5.25^{\text{th}} \text{ position} = 3.15,$$

$$Q_2 = 10.5^{\text{th}} \text{ position} = 4.1, \qquad Q_3 = 15.75^{\text{th}} \text{ position} = 4.8.$$

$$\text{IQR} = 4.8 - 3.15 = 1.65, \quad \text{LF} = 3.15 - 1.5(1.65), \quad \text{UF} = 4.8 + 1.5(1.65).$$

Worked Problem 11: Body Parameters of Asian Male Samples

Example 4.11. The following data represent variations in body parameters of 50 Asian male samples. Construct a stem-and-leaf plot and compute the measures of central tendency.

28	56	68	58	34	74	52	37	62	98
33	40	48	08	34	65	37	19	46	76
16	70	99	35	67	61	07	10	29	94
72	66	26	28	85	17	14	39	66	03
16	79	23	92	09	30	73	55	23	46

Ordered stem-and-leaf plot.

Stem	Ordered leaf
0	3 7 8 9
1	0 4 6 6 7 9
2	3 3 6 8 8 9
3	0 3 4 4 5 7 7 9
4	0 6 6 8
5	2 5 6 8
6	1 2 6 6 7 8
7	0 2 3 4 6 9
8	5
9	2 4 8 9

Measures of central tendency.

$$\text{Mean} = 45.46, \qquad \text{Median} = 39.5.$$

4.6 The Radar Chart

Radar chart

A **radar chart** is a graphical method of displaying multivariate data in the form of a two-dimensional chart of three or more quantitative variables represented on axes starting from the same point. Radar charts — also known as spider charts, polar charts, web charts, or star plots — are a way to visualise multivariate data. They are used to plot one or more groups of values over multiple common variables.

The radar chart consists of a sequence of equiangular spokes, called radii, with each spoke representing one of the variables. The data length of a spoke is proportional to the magnitude of the variable for the data point relative to the maximum magnitude of the variable across all data points. A line is drawn connecting the data values for each spoke. This gives the plot a star-like appearance, and the origin of one of its popular names.

A radar chart can be used to answer the following questions:

- Which observations are most similar; that is, are there clusters of observations? (Radar charts are used to examine the relative values for a single data point and to locate similar points or dissimilar points.)
- Are there outliers?

Developing radar charts.

1. Identify the alternatives to be compared.
2. Generate criteria to rate each alternative.
3. Rate each alternative based on the criteria.
4. Select a suitable structure and set relevant options; set and label the axes.
5. Label the alternative's ratings on the chart.

Guidelines.

1. No more than 15 alternatives should be chosen, or else the chart will look cramped.
2. Use at least 3 criteria, but more than 7 will become complex.
3. Standardise ratings to a uniform scale.

Example 4.12 (Manufacturing process ratings)**.** Three criteria are used to compare a process against a best-in-class benchmark.

Category	Our score	Best-in-class	Standardised
1. Safety	Red. 5 accidents	Red. 12 accidents	$5/12 = 42\%$
2. Training expense	Rs. 10,000	Rs. 20,000	50%
3. Standard operating procedures	74%	96%	$74/96 = 77\%$

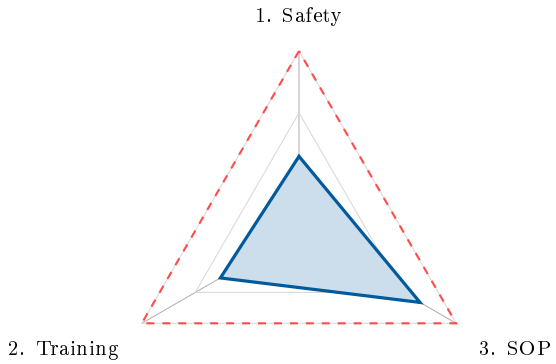


Figure 4.10: Radar chart comparing the process (solid) against the best-in-class benchmark (dashed), with each axis standardised to a common scale.

Chapter 5

Concepts of Probability and Random Variables

Probability provides the formal language for reasoning about uncertainty. This chapter introduces random experiments, sample spaces, events, the algebra of events, and the counting techniques that underlie the computation of probabilities for equally likely outcomes.

Engineering systems are usefully pictured as transforming inputs into outputs under the influence of two kinds of variables: **controlled variables**, which the engineer sets, and **noise variables**, which the engineer cannot control. It is the noise variables that introduce randomness into the output.

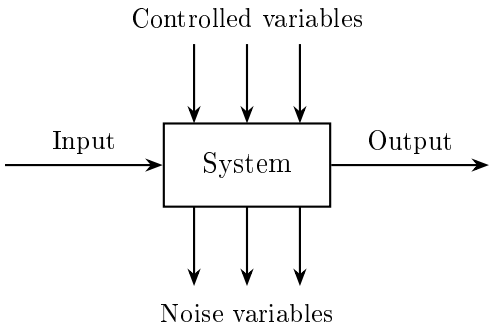


Figure 5.1: A system viewed as a transformation of inputs to outputs, subject to controlled and noise variables.

5.1 Random Experiments and Sample Spaces

Random experiment

A **random experiment** is an experiment that can result in different outcomes, even though it is repeated in the same manner every time.

Sample space

The set of all possible outcomes of a random experiment is called the **sample space** of the experiment. The sample space is denoted by S. A sample space is often defined based on the objectives of the analysis.

Example 5.1 (Thickness of a molded part)**.** Consider an experiment in which you select a molded plastic part, such as a connector, and measure its thickness. The possible values for thickness depend on the resolution of the measuring instrument, and also depend on upper and

lower bounds for thickness. It might be convenient to define the sample space as simply the positive real line, since a negative value for thickness cannot occur:

$$S = \mathbb{R}^+ = \{\, x \mid x > 0 \,\}.$$

If it is known that all connectors will be between 10 and 11 millimetres thick, the sample space could be

$$S = \{\, x \mid 10 < x < 11 \,\}.$$

If the objective of the analysis is to consider only whether a particular part is low, medium, or high for thickness, the sample space will be the set of three outcomes:

$$S = \{\text{low},\ \text{medium},\ \text{high}\}.$$

If the objective is to consider only whether or not a particular part conforms to the manufacturing specifications, the sample space is

$$S = \{\text{yes},\ \text{no}\},$$

indicating whether or not the part conforms.

Discrete and continuous sample spaces

A sample space is **discrete** if it consists of a finite or countably infinite set of outcomes. A sample space is **continuous** if it contains an interval (either finite or infinite) of real numbers.

In the previous examples, the choice $S = \mathbb{R}^+$ is an example of a continuous sample space, whereas $S = \{\text{yes}, \text{no}\}$ is a discrete sample space.

Example 5.2 (Sample spaces for various experiments)**.**

Experiment	Sample space
Throw a die	$\{1, 2, \ldots, 6\}$
Record claim amounts in all insurance policies	$\{0, 1, \ldots, \infty\}$
Record height of students (cm)	$\{\, x \mid 100 \leq x \leq 250 \,\}$
Record number of accidents per day in front of a gate	$\{0, 1, 2, \ldots, 100\}$

The sample spaces that contain the outcomes of dice rolls, drawings from a bag of mixed-colour balls, and dealings from a regular 52-card deck are examples of discrete sample spaces. Another example is the number of roulette-wheel spins made before the ball lands on 25; the number can range from $1, 2, 3, \ldots$ up to infinity, but the number has to be an integer. Sample spaces that contain the outcomes of temperature readings, height measurements, and salaries are examples of continuous sample spaces.

5.1.1 Sampling With and Without Replacement

In random experiments in which items are selected from a batch, we indicate whether or not the selected item is replaced before the next one is selected.

Example 5.3 (With and without replacement)**.** If the batch consists of 3 items $\{a, b, c\}$ and our experiment is to select 2 items *without* replacement, the sample space of ordered outcomes can be represented as

$$S_{\text{without}} = \{ab, ac, ba, bc, ca, cb\}, \quad \text{or unordered as } \{\{a, b\}, \{b, c\}, \{c, a\}\}.$$

If items are *replaced* before the next one is selected, the sampling is referred to as *with replacement*. Then the possible ordered outcomes are

$$S_{\text{with}} = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}.$$

Example 5.4 (Good and defective parts)**.** Suppose that there are 5 defective parts and 95 good parts in a batch. To study the quality of the batch, 2 are selected without replacement. Let g denote a good part and d denote a defective part. The ordered sample space in this case is

$$S = \{gg, gd, dg, dd\}.$$

If there is only one defective part in the batch, then the sample space is

$$S = \{gg, gd, dg\}.$$

5.1.2 Tree Diagrams

Sample spaces can also be described with tree diagrams. Consider a digital communication system in which each message is classified as to whether it is received within the time specified by the system design — either "on time" or "late". If 3 messages are classified, the possible results of the 3 messages can be displayed by 8 branches in the tree diagram.

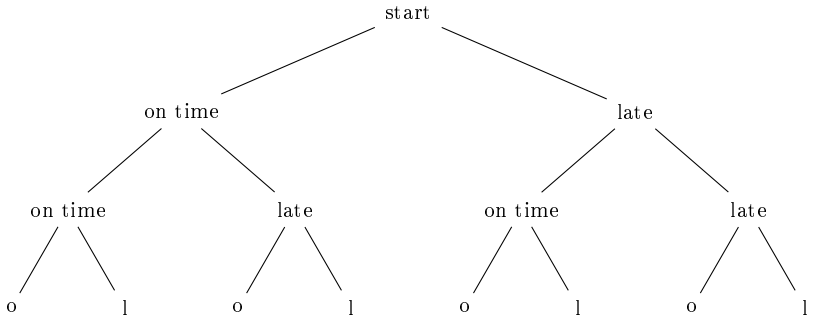


Figure 5.2: Tree diagram for the classification of 3 messages, each "on time" (o) or "late" (l), giving $2^3 = 8$ branches.

5.2 Events and the Algebra of Events

Event

An **event** is a subset of the sample space of a random experiment.

Union, intersection, complement

For two events E_1 and E_2:

- The **union** $E_1 \cup E_2$ is the event that consists of all outcomes that are contained in either of the two events.
- The **intersection** $E_1 \cap E_2$ is the event that consists of all outcomes that are contained in both of the two events.
- The **complement** of an event in the sample space is the set of outcomes in the sample space that are not in the event. It is denoted E' or E^c.

Example 5.5 (Events on a continuous sample space)**.** Measurements of the thickness of a plastic connector might be modelled with sample space $S = \mathbb{R}^+$, the set of positive real numbers. Let

$$E_1 = \{\, x \mid 10 \le x < 12 \,\}, \qquad E_2 = \{\, x \mid 11 < x < 15 \,\}.$$

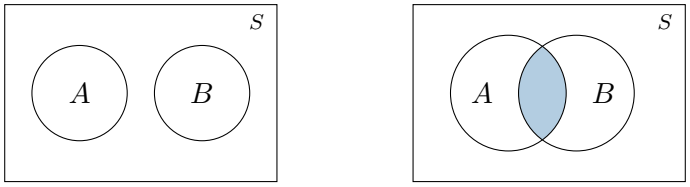
Then

$$E_1 \cup E_2 = \{\, x \mid 10 \le x < 15 \,\}, \qquad E_1 \cap E_2 = \{\, x \mid 11 < x < 12 \,\},$$
$$E_1' = \{\, x \mid x < 10 \text{ or } 12 \le x \,\}, \qquad E_1' \cap E_2 = \{\, x \mid 12 \le x < 15 \,\}.$$

Mutually exclusive events

Two events E_1 and E_2 such that $E_1 \cap E_2 = \varnothing$ are said to be **mutually exclusive**.

Venn diagrams are used to represent a sample space and events in the sample space.



Mutually exclusive events $A \cap B$

Figure 5.3: Venn diagrams: mutually exclusive events (left) and the intersection $A \cap B$ of two overlapping events (right).

5.2.1 Laws of Set Operations

Property 5.1 (Complement of a complement)**.**

$$(E')' = E.$$

Property 5.2 (Distributive laws)**.**

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C),$$
$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C).$$

Property 5.3 (De Morgan's laws)**.**

$$(A \cup B)' = A' \cap B', \qquad (A \cap B)' = A' \cup B'.$$

Property 5.4 (Commutativity)**.**

$$A \cap B = B \cap A, \qquad A \cup B = B \cup A.$$

5.3 Counting Techniques

Counting techniques determine the number of outcomes in each event, which is essential for computing probabilities when outcomes are equally likely.

Multiplication rule

If an operation can be described as a sequence of k steps, and the number of ways of completing the first step is n_1, and the number of ways of completing the second step is n_2 for each way of completing the first step, and so forth, then the total number of ways of completing the operation is

$$n_1 \times n_2 \times \cdots \times n_k.$$

5.3.1 Permutations

Permutation

A **permutation** is an ordered sequence of elements of a set. For a set $S = \{a, b, c\}$, the permutations are $abc, acb, bac, bca, cab, cba$. The number of permutations of n different elements is

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1.$$

Permutations of subsets

The number of permutations of subsets of r elements selected from a set of n different elements is

$$P_r^n = n \times (n-1) \times (n-2) \times \cdots \times (n-r+1) = \frac{n!}{(n-r)!}.$$

Example 5.6 (Circuit-board component placement)**.** A printed circuit board has 8 different locations in which a component can be placed. If four different components are to be placed on the board, how many different designs are possible? Using the permutation formula,

$$P_4^8 = 8 \times 7 \times 6 \times 5 = \frac{8!}{4!} = 1680 \text{ different designs.}$$

Permutations of similar objects

The number of permutations of $n = n_1 + n_2 + \cdots + n_r$ objects, of which n_1 are of one type, n_2 are of a second type, ..., and n_r are of an rth type, is

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}.$$

Example 5.7 (Part labels from lines)**.** A part is labelled by printing with 4 thick lines, 3 medium lines, and 2 thin lines. If each ordering of 9 lines represents a different label, how many different labels can be generated by using this scheme? The number of possible part labels is

$$\frac{9!}{4!\,3!\,2!} = 1260.$$

5.3.2 Combinations

Combination

The number of **combinations** — subsets of size r that can be selected from a set of n elements — is denoted $\binom{n}{r}$ or C_r^n, and

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$

Example 5.8 (Samples with exactly two defectives)**.** A bin of 50 parts manufactured contains 3 defective parts and 47 non-defective parts. A sample of six parts is selected from the 50 parts and not replaced. How many different samples are there of size six that contain exactly 2 defective parts?

A subset containing exactly 2 defective parts can be formed by choosing 2 defective parts from 3 defective parts:

$$\binom{3}{2} = \frac{3!}{1!\,2!} = 3 \text{ ways.}$$

The second step is to select the remaining 4 parts from the 47 acceptable parts in the bin:

$$\binom{47}{4} = \frac{47!}{4!\,43!} = 178365 \text{ different ways.}$$

By the multiplication rule, the number of subsets of size six that contain exactly 2 defective items is

$$3 \times 178365 = 535{,}095.$$

The total number of different subsets of size six is

$$\binom{50}{6} = \frac{50!}{44!\,6!} = 15{,}890{,}700.$$

Therefore the probability is

$$\text{probability} = \frac{535{,}095}{15{,}890{,}700} = 0.034 = \frac{\text{number of outcomes in event}}{\text{number of outcomes in sample space}}.$$

5.4 Sample-Space Examples

Example 5.9 (Three machined parts above/below target)**.** Each of 3 machined parts is classified as either above or below the target specification for the part. Let a and b denote a part above and below specification respectively. Then

$$S = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}.$$

Example 5.10 (Nonconformities in power supplies)**.** In the final inspection of electronic power supplies, 3 types of nonconformities might occur: functional, minor, or cosmetic. Power supplies that are defective are further classified as to the type of nonconformity. With a denoting an acceptable power supply and f, m, c the defective categories,

$$S = \{a, f, m, c\}.$$

Example 5.11 (Three-digit ammeter)**.** An ammeter that displays 3 digits is used to measure current in milliamperes. The sample space is the 1000 possible 3-digit integers:

$$S = \{000, 001, \ldots, 999\}.$$

Example 5.12 (pH reading). The pH reading of a water sample to the nearest tenth of a unit has sample space

$$S = \{1.0, 1.1, 1.2, 1.3, \ldots, 14.0\}.$$

Example 5.13 (Repeated calls to a busy line). Calls are repeatedly placed to a busy phone line until a connect is achieved. With c denoting connect and b denoting busy,

$$S = \{c, bc, bbc, bbbc, bbbbc, \ldots\}.$$

Example 5.14 (Length and width in injection molding). In an injection-molding operation, the length and width, denoted X and Y respectively, of each molded part are evaluated. Let A denote the event $48 < X < 52$ cm; let B denote the event $9 < Y < 11$ cm; and let C denote the event that a critical length meets customer requirements. If the events were mutually exclusive, then $A \cap B$ would be the null set, so the process could not produce parts with $X = 50$ cm and $Y = 10$ cm; such a process would not be successful.

Example 5.15 (Sampling printed circuit boards without replacement). A sample of 2 printed circuit boards is selected without replacement from a batch. Let $\bar{d}$ denote no defect, d denote a minor defect, and D denote a major defect.

- If the batch contains 90 boards that are not defective, 8 boards with minor defects, and 2 boards with major defects, the ordered sample space is

$$S = \{\bar{d}\bar{d},\ \bar{d}d,\ \bar{d}D,\ d\bar{d},\ dd,\ dD,\ D\bar{d},\ Dd,\ DD\}.$$

- If the batch contains 90 boards not defective, 8 boards with minor defects, and 1 board with a major defect, the sample space is similarly $S = \{\bar{d}\bar{d}, \bar{d}D, \bar{d}d, d\bar{d}, dd, dD, D\bar{d}, Dd\}$.

Example 5.16 (Configuring a personal digital assistant). An order for a personal digital assistant can specify any one of 5 memory sizes, any one of 3 types of displays, any one of 4 sizes of hard disk, and can either include or not include a pen tablet. How many different systems can be ordered? By the multiplication rule,

$$5 \times 3 \times 4 \times 2 = 120 \text{ different systems.}$$

Example 5.17 (Production sequences). A manufacturing process consists of 10 operations that can be completed in any order. How many different production sequences are possible? Since once an operation is selected it cannot be selected again,

$$10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 3{,}628{,}800.$$

Example 5.18 (Inspecting semiconductor chips). A lot of 140 semiconductor chips is inspected by choosing a sample of five chips. Assume 10 of the chips do not conform to customer requirements.

(a) **How many different samples are possible?**

$$\binom{140}{5} = \frac{140!}{135!\,5!} = 41{,}69{,}65{,}528.$$

(b) **How many samples of 5 contain exactly one nonconforming chip?**

$$\binom{130}{4}\binom{10}{1} = \frac{130!}{126!\,4!} \times \frac{10!}{9!\,1!} = 11{,}35{,}88{,}800.$$

(c) **How many samples of 5 contain at least one nonconforming chip?** Counting the disjoint cases (4 good and 1 bad), (3 good and 2 bad), (2 good and 3 bad), (1 good and 4 bad), and (0 good and 5 bad):

$$\binom{130}{4}\binom{10}{1} + \binom{130}{3}\binom{10}{2} + \binom{130}{2}\binom{10}{3} + \binom{130}{1}\binom{10}{4} + \binom{130}{0}\binom{10}{5}$$
$$= 113588800 + 16099200 + 1006200 + 27300 + 252 = 130{,}721{,}752.$$

Example 5.19 (Stacking components in a cylindrical casing)**.** In the design of an electromechanical product, 7 different components are to be stacked into a cylindrical casing that holds 12 components in a manner that minimises the impact of shocks. One end of the casing is designated as the bottom and the other end is the top.

(a) **How many different designs are possible?**

$$P_7^{12} = \binom{12}{7} \times 7! = \frac{12!}{7!\,5!} \times 7! = \frac{12!}{5!} = 3{,}99{,}16{,}800.$$

(b) **If the seven components are all identical, how many different designs are possible?** Since 7 of 12 positions are occupied by identical components (equivalently $n_6 = 7$ identical and the rest distinct positions),

$$\frac{12!}{5!\,7!} = \binom{12}{7} = 95{,}040 \quad \text{(choosing positions)},$$

more precisely $\dfrac{12!}{1!\,1!\,1!\,1!\,1!\,7!} = 95{,}040$ when 5 positions are distinct and 7 are identical.

(c) **If the 7 components consist of 5 of one type and 4 of another type**[1]**, how many different designs are possible?**

$$\binom{12}{7} \times \frac{7!}{3!\,4!} = \frac{12!}{5!\,3!\,4!} = 27{,}720.$$

[1] As recorded in the source; here the multiset arrangement formula is applied.

Chapter 6

Axioms of Probability and the Addition Rules

Having established how to count outcomes, we now assign probabilities to events and develop the rules that govern how probabilities combine.

6.1 Equally Likely Outcomes

Equally likely outcomes

Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.

For a discrete sample space, the probability of an event E, denoted $\mathrm{P}(E)$, equals the sum of the probabilities of the outcomes in E.

6.2 The Axioms of Probability

Axioms of probability

Probability is a number that is assigned to each member of a collection of events from a random experiment that satisfies the following properties. If S is the sample space and E is any event in a random experiment, then:
(1) $\mathrm{P}(S) = 1$.
(2) $0 \leq \mathrm{P}(E) \leq 1$.
(3) For two events E_1 and E_2 with $E_1 \cap E_2 = \varnothing$,

$$\mathrm{P}(E_1 \cup E_2) = \mathrm{P}(E_1) + \mathrm{P}(E_2).$$

Two immediate consequences are

$$\mathrm{P}(\varnothing) = 0, \qquad \mathrm{P}(E') = 1 - \mathrm{P}(E) \text{ for any event } E.$$

6.3 The Addition Rule

Addition rule for two events

For any two events A and B,

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B).$$

This follows from a decomposition of the union into disjoint pieces. Writing $\mathrm{P}(A) = x + y$ and $\mathrm{P}(B) = y + z$ where $y = \mathrm{P}(A \cap B)$, the union covers $x + y + z = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$.

Property 6.1 (Mutually exclusive events)**.** *If A and B are mutually exclusive, then* $\mathrm{P}(A \cap B) = 0$ *and*

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B).$$

More generally, if $E_1, E_2, \ldots, E_k$ is a collection of mutually exclusive events — so that $E_i \cap E_j = \varnothing$ for all pairs — then

$$\mathrm{P}(E_1 \cup E_2 \cup \cdots \cup E_k) = \mathrm{P}(E_1) + \mathrm{P}(E_2) + \cdots + \mathrm{P}(E_k).$$

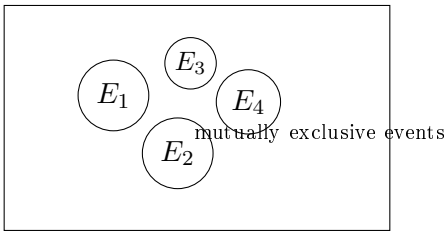


Figure 6.1: A collection of mutually exclusive events partitioning part of the sample space.

Addition rule for three events

$$\begin{aligned}\mathrm{P}(A \cup B \cup C) &= \mathrm{P}\big[(A \cup B) \cup C\big] = \mathrm{P}(A \cup B) + \mathrm{P}(C) - \mathrm{P}\big[(A \cup B) \cap C\big] \\ &= \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(A \cap B) - \mathrm{P}(A \cap C) \\ &\quad - \mathrm{P}(B \cap C) + \mathrm{P}(A \cap B \cap C).\end{aligned}$$

6.4 Worked Problems on Probability of Events

Venn Diagram Problem: Length and Width

Example 6.1. In an injection-molding operation, the length and width X and Y of each molded part are evaluated. Let A denote $48 < X < 52$ cm, B denote $9 < Y < 11$ cm, and C denote that a critical length meets customer requirements. Construct a Venn diagram that includes these events. Shade the areas that represent (a) A; (b) $A \cap B$; (c) $A' \cup B$; (d) $A \cup B$. (e) If these events were mutually exclusive, how successful would this production operation be? Would the process produce parts with $X = 50$ cm and $Y = 10$ cm?

(e) If the events are mutually exclusive, then $A \cap B$ is the null set. Therefore the process does not produce parts with $X = 50$ cm and $Y = 10$ cm, and the process would not be successful.

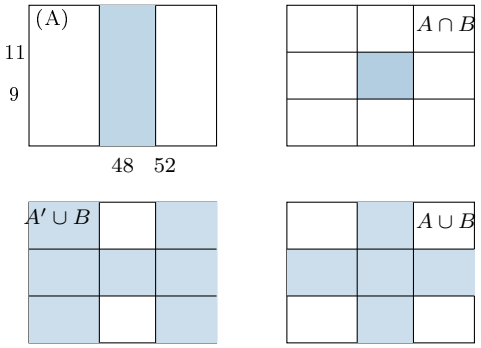


Figure 6.2: Shaded Venn-style regions for the length–width events: A, $A \cap B$, $A' \cup B$, and $A \cup B$, where A is the vertical band $48 < X < 52$ and B is the horizontal band $9 < Y < 11$.

Conformance of Emissions Samples

Example 6.2. Samples of emissions from 3 suppliers are classified for conformance to air-quality specifications. The results from 100 samples are summarised below.

Supplier	Conforms: Yes	No
1	22	8
2	25	5
3	30	10

Let A denote the event that a sample is from supplier 1, and let B denote the event that a sample conforms to specifications. Determine the number of samples in $A' \cap B$, B', and $A \cup B$.

Solution.

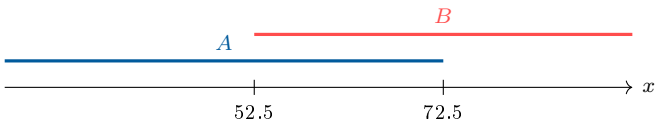
$$\begin{aligned} A' \cap B &= \text{sample not from supplier 1 and conforms} = 25 + 30 = 55, \\ B' &= \text{sample not conforming} = 8 + 5 + 10 = 23, \\ A \cup B &= \text{samples from supplier 1, or conforming} \\ &= (22 + 8) + (22 + 25 + 30) - 22 = 85. \end{aligned}$$

Reactor Rise Time

Example 6.3. The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be the positive real numbers. Events A and B are defined as

$$A = \{\, x \mid x < 72.5 \,\}, \qquad B = \{\, x \mid x > 52.5 \,\}.$$

Describe each of the following events: (a) A'; (b) B'; (c) $A \cap B$; (d) $A \cup B$.



Solution.

$$A \cup B = S = \{\, x \mid 0 < x < \infty \,\}, \qquad A \cap B = S' = \{\, x \mid 52.5 < x < 72.5 \,\}.$$

(Here A' is $\{x \mid x \geq 72.5\}$ and B' is $\{x \mid x \leq 52.5\}$.)

Last Digit of a Weight Measurement

Example 6.4. If the last digit of a weight measurement is equally likely to be any of the digits 0 through 9: (a) What is the probability that the last digit is 0? (b) What is the probability that the last digit is greater than or equal to 5?

$$\text{(a)}\ \ \frac{1}{10} = 0.1, \qquad \text{(b)}\ \ \frac{5}{10} = 0.5.$$

Validity of a Credit-Card Number

Example 6.5. A credit card contains 16 digits between 0 and 9. However, only 100 million numbers are valid. If a number is entered randomly, what is the probability that it is a valid number?

The total number of possibilities is

$$N_T = (\text{number of states for each digit})^{\text{number of digits}} = 10^{16}.$$

Let A represent the event that the random number entered is a valid number, so $n(A) = 100 \times 10^6$. Then

$$\text{probability} = \frac{n(A)}{N} = \frac{100 \times 10^6}{10^{16}} = \frac{1}{10^8} = 1 \times 10^{-8}.$$

Licence-Plate Selection

Example 6.6. Suppose your vehicle is licensed in a state that issues license plates consisting of 3 digits (between 0 and 9) followed by 3 letters (between A and Z). If a license number is selected randomly, what is the probability that yours is the one selected?

$$N_D = 10^3, \qquad N_A = 26^3, \qquad \text{probability} = \frac{1}{10^3 \times 26^3}.$$

A Four-Event Probability Table

Example 6.7. For events A and B with the supplier data ($\mathrm{P}(A) = 30/100$, and a second event B with 77 favourable outcomes), compute (a) $\mathrm{P}(A)$; (b) $\mathrm{P}(B)$; (c) $\mathrm{P}(A')$; (d) $\mathrm{P}(A \cap B)$; (e) $\mathrm{P}(A \cup B)$; (f) $\mathrm{P}(A' \cup B)$.

(a) $\mathrm{P}(A) = \frac{30}{100} = 0.3$, (b) $\mathrm{P}(B) = \frac{77}{100} = 0.77$,
(c) $\mathrm{P}(A') = 0.7$, (d) $\mathrm{P}(A \cap B) = \frac{22}{100} = 0.22$,
(e) $\mathrm{P}(A \cup B) = 0.30 + 0.77 - 0.22 = \frac{85}{100} = 0.85$, (f) $\mathrm{P}(A' \cup B) = 0.70 + 0.77 - 0.55 = \frac{92}{100} = 0.92$.

Polycarbonate Disks: Scratch and Shock Resistance

Example 6.8. Disks of polycarbonate plastic from a supplier are analysed for scratch and shock resistance. The results from 100 disks are summarised below.

		Shock resistance	
		High	Low
Scratch resistance	High	70	9
	Low	16	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. If a disk is selected at random, determine (a) $\mathrm{P}(A)$; (b) $\mathrm{P}(B)$; (c) $\mathrm{P}(A')$; (d) $\mathrm{P}(A \cap B)$; (e) $\mathrm{P}(A \cup B)$; (f) $\mathrm{P}(A' \cup B)$.

$$\text{(a) } \mathrm{P}(A) = \tfrac{86}{100} = 0.86, \qquad \text{(b) } \mathrm{P}(B) = 0.79,$$
$$\text{(c) } \mathrm{P}(A') = \tfrac{14}{100} = 0.14, \qquad \text{(d) } \mathrm{P}(A \cap B) = 0.70,$$
$$\text{(e) } \mathrm{P}(A \cup B) = 0.86 + 0.79 - 0.70 = 0.95, \qquad \text{(f) } \mathrm{P}(A' \cup B) = 0.14 + 0.79 - 0.09 = 0.84.$$

Customer Orders of Items A, B, C

Example 6.9. Customers order any combination of items A, B, C. The following probabilities are given:

$$\mathrm{P}(C) = 0.3, \quad \mathrm{P}(B) = 0.4, \quad \mathrm{P}(A \cap B) = 0.15, \quad \mathrm{P}(A \cap C) = 0.15,$$
$$\mathrm{P}(B \cap C) = 0.55, \quad \mathrm{P}(A \cap B \cap C) = 0.05, \quad \mathrm{P}(\text{none}) = 0.2.$$

Find $\mathrm{P}(A)$ and the probability that A alone is ordered.

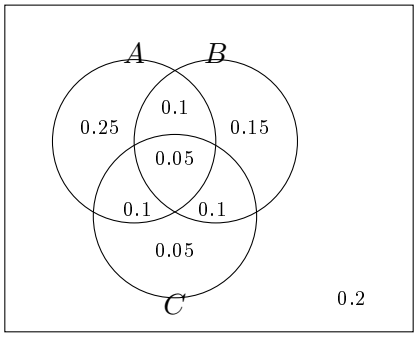


Figure 6.3: Venn diagram for the customer-order problem, with probabilities assigned to each disjoint region.

Solution. First, using the addition rule on B and C,

$$\mathrm{P}(B \cup C) = \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(B \cap C) \Rightarrow 0.55 = 0.4 + 0.3 - \mathrm{P}(B \cap C),$$

so $\mathrm{P}(B \cap C) = 0.7 - 0.55 = 0.15$ (the probability that B and C are both ordered). Then

$$\mathrm{P}(A \text{ alone}) = 1 - 0.2 - (0.3 + 0.15 + 0.1) = 0.25,$$

and assembling the regions of A,

$$\mathrm{P}(A) = 0.25 + 0.1 + 0.1 + 0.05 = 0.5.$$

Automobile Headlamp Performance

Example 6.10. A manufacturer of front lights for automobiles tests lamps under a high-humidity, high-temperature environment using intensity and useful life as the responses of interest. The following table shows the performance of 130 lamps.

		Useful life	
		Satisfactory	Unsatisfactory
Intensity	Satisfactory	117	3
	Unsatisfactory	8	2

(i) Find the probability that a randomly selected lamp will yield unsatisfactory results under any criterion. (ii) The customers for these lamps demand 95% satisfactory results. Can the lamp manufacturer meet this demand?

Solution.

$$\text{(i)}\quad \text{P(unsatisfactory under some criterion)} = \frac{3+2+8}{130} = \frac{13}{130} = 0.1.$$

$$\text{(ii)}\quad \text{Satisfactory under both criteria} = \frac{117}{130} = 90\%.$$

Since only 90% are satisfactory, the lamp manufacturer has *not* met the 95% demand.

Chapter 7

Conditional Probability, Independence, and Bayes' Theorem

Often the probability of an event changes once we learn that another event has occurred. Conditional probability formalises this updating of belief, and leads to the multiplication rule, the total probability rule, the notion of independence, and finally Bayes' theorem.

7.1 Conditional Probability

Conditional probability

The **conditional probability** of an event B given an event A, denoted $\mathrm{P}(B \mid A)$, is

$$\mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)}, \qquad \text{for } \mathrm{P}(A) > 0.$$

This definition can be understood in the special case in which all outcomes of a random experiment are equally likely. If there are n total outcomes,

$$\mathrm{P}(A) = \frac{\text{number of outcomes in } A}{n}, \qquad \mathrm{P}(A \cap B) = \frac{\text{number of outcomes in } A \cap B}{n},$$

so that

$$\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{\text{number of outcomes in } A \cap B}{\text{number of outcomes in } A}.$$

Defective Parts Without Replacement

Example 7.1. A day's production of 850 manufactured parts contains 50 parts that do not meet customer requirements. Two parts are selected randomly without replacement from the batch. What is the probability that the second part is defective given that the first part is defective?

Let A denote the event that the first part is defective and B the event that the second part is defective. After one defective part is removed, 49 defectives remain among 849 parts, so

$$\mathrm{P}(B \mid A) = \frac{49}{849}.$$

Example 7.2 (Three parts without replacement)**.** If 3 parts are selected randomly from the same batch, what is the probability that the first two are defective and the third is not defective?

$$\mathrm{P}(d\,d\,n) = \frac{50}{850} \times \frac{49}{849} \times \frac{800}{848} = 0.0032.$$

Conformance and Suppliers (Continued)

Example 7.3. Using the supplier conformance data, with $\mathrm{P}(A) = 0.86$ and $\mathrm{P}(B) = 0.79$ and $\mathrm{P}(A \cap B) = 0.70$:

$$\mathrm{P}(A \mid B) = \frac{0.70}{0.79} = 0.886, \qquad \mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{0.70}{0.86} = 0.814.$$

Air-Conditioning Failure Mechanisms

Example 7.4. A maintenance firm has gathered the following information regarding the failure mechanisms for air-conditioning systems.

		Evidence of gas leaks	
		Yes	No
Evidence of electrical failure	Yes	55	17
	No	32	3

The units without evidence of gas leaks or electrical failure showed other types of failure. If this is a representative sample of AC failure, find the probability that (a) failure involves a gas leak; (b) there is evidence of electrical failure given that there was a gas leak; (c) there is evidence of a gas leak given that there is evidence of electrical failure.

Solution. Let E represent electrical failure and G represent a gas leak. The marginal totals give

$$\mathrm{P}(E) = \frac{72}{107} = 0.67, \quad \mathrm{P}(E') = \frac{35}{107} = 0.33, \quad \mathrm{P}(G) = \frac{87}{107} = 0.81.$$

$$\begin{aligned}
&\text{(i)} \quad \mathrm{P}(G) = \frac{87}{107} = 0.81, \\
&\text{(ii)} \quad \mathrm{P}(E \mid G) = \frac{\mathrm{P}(E \cap G)}{\mathrm{P}(G)} = \frac{55/107}{87/107} = \frac{55}{87}, \\
&\text{(iii)} \quad \mathrm{P}(G \mid E) = \frac{\mathrm{P}(G \cap E)}{\mathrm{P}(E)} = \frac{55}{72}.
\end{aligned}$$

Cast-Aluminium Parts: Surface Finish and Length

Example 7.5. Samples of a cast-aluminium part are classified on the basis of surface finish (in microinches) and length measurements. The results of 100 parts are summarised below.

		Length	
		Excellent	Good
Surface finish	Excellent	80	2
	Good	10	8

Let A denote the event that the sample has excellent surface finish, and B the event that the sample has excellent length. Find (a) $\mathrm{P}(A)$; (b) $\mathrm{P}(B)$; (c) $\mathrm{P}(A \mid B)$; (d) $\mathrm{P}(B \mid A)$; (e) if a selected part has excellent surface finish, the probability that the length is excellent; (f) if a selected part has good length, the probability that the surface finish is excellent.

Solution.

$$\mathrm{P}(A) = \frac{82}{100} = 0.82, \qquad \mathrm{P}(B) = \frac{90}{100} = 0.90,$$
$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{80}{90} = 0.889, \qquad \mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{80}{82} = 0.976,$$
$$\mathrm{P}(B \mid A) = 0.976, \qquad \mathrm{P}(A \mid B') = \frac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} = \frac{2}{10} = 0.2.$$

Melanin and Moisture Content

Example 7.6. With $A \to$ low melanin content and $B \to$ high moisture content, the joint counts are:

		Melanin content	
		High	Low
Moisture content	High	13	7
	Low	48	32

Then

$$\mathrm{P}(A) = \frac{39}{100} = 0.39, \quad \mathrm{P}(B) = \frac{20}{100} = 0.2,$$
$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(B \cap A)}{\mathrm{P}(B)} = \frac{7}{20} = 0.35, \qquad \mathrm{P}(B \mid A) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)} = \frac{7}{39}.$$

Deceased Beetles: Autolysis and Putrefaction

Example 7.7. The following table summarises the number of deceased beetles under autolysis (the destruction of a cell after its death by the action of its own enzymes) and putrefaction (decomposition of organic matter, especially protein, by microorganisms, resulting in foul-smelling matter).

		Autolysis	
		High	Low
Putrefaction	High	14	59
	Low	18	9

(a) If the autolysis of a sample is high, what is the probability that putrefaction is low? (b) If the putrefaction of a sample is high, what is the probability that autolysis is high? (c) If the putrefaction of a sample is low, what is the probability that autolysis is low?

Solution. Let $A \to$ autolysis high and $P \to$ putrefaction high. The totals give

$$\mathrm{P}(A) = \frac{32}{100}, \quad \mathrm{P}(A') = \frac{68}{100}, \quad \mathrm{P}(P) = \frac{73}{100}, \quad \mathrm{P}(P') = \frac{27}{100}.$$

$$\text{(a)} \quad \mathrm{P}(P' \mid A) = \frac{\mathrm{P}(P' \cap A)}{\mathrm{P}(A)} = \frac{18}{32} = 0.5625,$$
$$\text{(b)} \quad \mathrm{P}(A \mid P) = \frac{\mathrm{P}(A \cap P)}{\mathrm{P}(P)} = \frac{14}{73} = 0.1918,$$
$$\text{(c)} \quad \mathrm{P}(A' \mid P') = \frac{\mathrm{P}(A' \cap P')}{\mathrm{P}(P')} = \frac{9}{27} = 0.333.$$

Semiconductor Chips Without Replacement

Example 7.8. A lot of 100 semiconductor chips contains 20 that are defective. Two are selected randomly, without replacement, from the lot. (a) What is the probability that the first one selected is defective? (b) What is the probability that the second one selected is defective given that the first one is defective? (c) What is the probability that both are defective? (d) How does the answer to part (b) change if chips selected were replaced prior to the next selection?

Solution.

$$\text{(a)}\quad \frac{20}{100} = 0.2,$$

$$\text{(b)}\quad \mathrm{P}(\text{2nd def} \mid \text{1st def}) = \frac{\text{number of defective items remaining}}{\text{total remaining}} = \frac{19}{99},$$

$$\text{(c)}\quad \mathrm{P} = \frac{20}{100} \times \frac{19}{99} = 0.03838,$$

$$\text{(d)}\quad \mathrm{P} = \frac{20}{100} = 0.2 \quad \text{(if the defective chip is replaced).}$$

Passwords: Vowel and Even-Number Constraints

Example 7.9. A password consists of exactly seven characters, where each character is one of 26 letters (a–z) or 10 integers (0–9). Let A denote the subset of passwords that begin with a vowel and let B denote the subset of passwords that end with an even number. (a) Suppose a hacker selects a password at random. What is the probability that your password is selected? (b) Suppose a hacker knows your password is in event A and selects a password at random from this subset. What is the probability that your password is selected? (c) Suppose a hacker knows your password is in $A \cap B$ and selects a password at random from this subset. Find the probability.

Solution. Each of the seven positions can be filled in 36 ways, so there are 36^7 equally likely passwords.

$$\text{(a)}\quad \mathrm{P}(\text{mine}) = \frac{1}{36^7},$$

$$\text{(b)}\quad N_A = 5 \times 36^6 = \frac{5 \times 36^7}{36}, \quad \text{so} \quad \mathrm{P}(\text{mine} \mid A) = \frac{1}{5 \times 36^6},$$

$$\text{(c)}\quad N_C = 5 \times 5 \times 36^5, \quad \text{so} \quad \mathrm{P}(\text{mine} \mid A \cap B) = \frac{1}{N_C} = \frac{1}{25 \times 36^5}.$$

There are 5 vowels for the first position and 5 even digits $(0, 2, 4, 6, 8)$ for the last position.

7.2 The Multiplication and Total Probability Rules

Multiplication Rule

The definition of conditional probability can be rewritten to give a general expression for the probability of the intersection of two events. This is called the *multiplication rule*:

$$\mathrm{P}(A \cap B) = \mathrm{P}(B \mid A)\,\mathrm{P}(A) = \mathrm{P}(A \mid B)\,\mathrm{P}(B).$$

Total Probability Rule

For any two events A and B,

$$\mathrm{P}(B) = \mathrm{P}(B \cap A) + \mathrm{P}(B \cap A') = \mathrm{P}(B \mid A)\,\mathrm{P}(A) + \mathrm{P}(B \mid A')\,\mathrm{P}(A').$$

Total probability rule (multiple events). Assume $E_1, E_2, \ldots, E_k$ are k mutually exclusive and exhaustive sets. Then

$$\mathrm{P}(B) = \mathrm{P}(B \cap E_1) + \mathrm{P}(B \cap E_2) + \cdots + \mathrm{P}(B \cap E_k)$$

$$= \mathrm{P}(B \mid E_1)\,\mathrm{P}(E_1) + \mathrm{P}(B \mid E_2)\,\mathrm{P}(E_2) + \cdots + \mathrm{P}(B \mid E_k)\,\mathrm{P}(E_k).$$

Electrical Connector Kept Dry or Wet

Example 7.10. The probability is 1% that an electrical connector that is kept dry fails during the warranty period of a portable computer. If a connector is ever wet, the probability of a failure during the warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during the warranty period?

Solution. Let B be the event that a connector fails. Let A be the event of a wet condition and A' the dry condition, with $\mathrm{P}(A) = 0.1$, $\mathrm{P}(A') = 0.9$, $\mathrm{P}(B \mid A) = 0.05$, and $\mathrm{P}(B \mid A') = 0.01$. By the total probability rule,

$$\mathrm{P}(B) = \mathrm{P}(B \mid A)\,\mathrm{P}(A) + \mathrm{P}(B \mid A')\,\mathrm{P}(A') = 0.05 \times 0.1 + 0.01 \times 0.9 = 0.005 + 0.009 = 0.014.$$

Edge Roughness of Slit Paper Products

Example 7.11. The edge roughness of slit paper products increases as knife blades wear. Only 1% of products slit with new blades have rough edges, 3% of products slit with blades of average sharpness exhibit roughness, and 5% of products slit with worn blades exhibit roughness. If 25% of the blades in manufacturing are new, 60% are of average sharpness, and 15% are worn, what is the proportion of products that exhibit edge roughness?

Solution. Let B be the event that a product exhibits edge roughness, and let A_1, A_2, A_3 denote new, average, and worn blades respectively, with $\mathrm{P}(A_1) = 0.01$, $\mathrm{P}(A_2) = 0.03$, $\mathrm{P}(A_3) = 0.05$ as conditional roughness probabilities and blade proportions 0.25, 0.60, 0.15. Then

$$\begin{aligned}\mathrm{P}(B) &= \mathrm{P}(B \mid A_1)\,\mathrm{P}(A_1) + \mathrm{P}(B \mid A_2)\,\mathrm{P}(A_2) + \mathrm{P}(B \mid A_3)\,\mathrm{P}(A_3) \\ &= 0.01 \times 0.25 + 0.03 \times 0.60 + 0.05 \times 0.15 = 0.028.\end{aligned}$$

Election Polling

Example 7.12. Let B be the event that a respondent voted for a particular candidate. If among 2020 respondents the probability of voting is $\mathrm{P}(B) = 0.62 \times 0.50 + 0.38 \times 0.53 = 0.5114$, the probability that a single specified respondent is the one selected is

$$\frac{1}{1033} = 0.00096.$$

Computer Keyboard Failures

Example 7.13. Computer keyboard failures are due to faulty electrical connects (12%) or mechanical defects (88%). Mechanical defects are related to loose keys (27%) or improper assembly (73%). Electrical connect defects are caused by defective wires (35%), improper connections (13%), or poorly welded wires (52%). (a) Find the probability that a failure is due to loose keys. (b) Find the probability that a failure is due to improperly connected or poorly welded wires.

Solution. Let E be electrical-connect failures and M mechanical defects, with $L \to$ loose keys and $A \to$ improper assembly, while $E \to$ defective wires + improper connection + poorly welded.

$$\text{(a)}\quad \mathrm{P}(L) = 0.27 \times 0.88 = 0.2376,$$
$$\text{(b)}\quad \mathrm{P}(\text{improper conn.} + \text{poor weld}) = 0.52 \times 0.12 + 0.13 \times 0.12 = 0.0078.$$

Heart Failures

Example 7.14. Heart failures are due to either natural occurrences (87%) or outside factors (13%). Outside factors are related to induced substances (73%) or foreign objects (27%). Natural occurrences are caused by arterial blockage (56%), disease (27%), and infection (17%). (a) Determine the probability that a failure is due to an induced substance. (b) Determine the probability that a failure is due to disease or infection.

Solution.

$$\text{(a)}\quad \mathrm{P}(I.S) = 0.73 \times 0.13 = 0.0949,$$
$$\text{(b)}\quad \mathrm{P}(D \text{ or } I) = 0.27 \times 0.87 + 0.17 \times 0.87 = 0.3828.$$

7.3 Independence

Two events are *independent* if any one of the following equivalent statements is true:

1. $\mathrm{P}(A \mid B) = \mathrm{P}(A)$,
2. $\mathrm{P}(B \mid A) = \mathrm{P}(B)$,
3. $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$.

Laboratory Sample Contamination

Example 7.15. The probability that a lab specimen contains high levels of contamination is 0.10. Five samples are checked and the samples are independent. (a) What is the probability that none contains high levels of contamination? (b) What is the probability that exactly one contains high levels of contamination? (c) What is the probability that at least one contains high levels of contamination?

Solution. Let A_i be the event that the ith lab specimen contains high contamination, with $\mathrm{P}(A_i) = 0.1$ and $\mathrm{P}(A_i') = 0.9$.

(a) By independence,

$$\mathrm{P}(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5') = 0.9 \times 0.9 \times 0.9 \times 0.9 \times 0.9 = 0.5905.$$

(c) The complement of "none" is "at least one":

$$1 - \mathrm{P}(A_1' \cap A_2' \cap A_3' \cap A_4' \cap A_5') = 1 - 0.5905 = 0.4095 \approx 0.41.$$

(b) The event that exactly one contains high levels of contamination is a disjoint union. For each $i_0 = 1, 2, 3, 4, 5$, the event that only specimen i_0 is contaminated is

$$\Big(\bigcap_{i \neq i_0} A_i'\Big) \cap A_{i_0}.$$

For example, for $i_0 = 2$,

$$\mathrm{P}\Big(\bigcap_{i \neq 2} A_i' \cap A_2\Big) = \mathrm{P}(A_1')\,\mathrm{P}(A_3')\,\mathrm{P}(A_4')\,\mathrm{P}(A_5')\,\mathrm{P}(A_2) = 0.9 \times 0.9 \times 0.9 \times 0.9 \times 0.1 = 0.066.$$

Since the five such events are disjoint and each has probability 0.066,

$$\mathrm{P}(E_1) = 0.066 + 0.066 + 0.066 + 0.066 + 0.066 = 0.33.$$

Acid–Base Titration by Two Technicians

Example 7.16. In an acid–base titration, a base or acid is gradually added to the other until they have completely neutralised each other. Since acids and bases are usually colourless, pH is measured to monitor the reaction. Suppose that the equivalence point is reached after approximately 100 mL of a NaOH solution has been added but that replicates are equally likely to indicate from 95 to 104 mL, measured to the nearest mL. Assume that two technicians each conduct titrations independently. (a) What is the probability that both technicians obtain equivalence at 100 mL? (b) What is the probability that both technicians obtain equivalence between 98 and 104 mL (inclusive)? (c) What is the probability that the average volume at equivalence from the two technicians is 100 mL?

Solution. Technicians T_1 and T_2 independently obtain equivalence at one of the quantities

$$Q = \{95, 96, 97, 98, 99, 100, 101, 102, 103, 104\} = \{q_i\}_{i=1}^{10},$$

with equal probability. Let t_1 and t_2 represent the quantities at which T_1 and T_2 obtain equivalence respectively, so $\mathrm{P}(t_1 = q_i) = \frac{1}{10}$ and $\mathrm{P}(t_2 = q_i) = \frac{1}{10}$.

(a)

$$\mathrm{P}\big(\{t_1 = 100\} \cap \{t_2 = 100\}\big) = \mathrm{P}(t_1 = 100)\,\mathrm{P}(t_2 = 100) = \frac{1}{10} \times \frac{1}{10} = \frac{1}{100}.$$

(b) There are 7 values in $\{98, \ldots, 104\}$, so

$$\mathrm{P}\big(t_1, t_2 \in \{98, \ldots, 104\}\big) = \mathrm{P}(t_1 \in \{98, \ldots, 104\})\,\mathrm{P}(t_2 \in \{98, \ldots, 104\}) = \frac{7}{10} \times \frac{7}{10} = \frac{49}{100}.$$

(c) We need pairs (t_1, t_2) such that $\frac{t_1 + t_2}{2} = 100$, i.e. $t_1 + t_2 = 200$. The favourable pairs are

$$(100, 100),\ (101, 99),\ (99, 101),\ (102, 98),\ (98, 102),\ (103, 97),\ (97, 103),\ (104, 96),\ (96, 104),$$

that is 9 pairs out of 100 equally likely outcomes. Hence the probability that the average volume at equivalence is 100 mL is

$$\frac{9}{100} = 0.09.$$

Spider-Mite Infestation of Ivy Plants

Example 7.17. Six tissues are extracted from an ivy plant infested by spider mites. The plant is infested in 20% of its area. A sample is chosen from randomly selected areas on the ivy plant. Assume that the samples are independent. (a) What is the probability that four successive samples show the signs of infestation? (b) What is the probability that 3 out of 4 successive samples show the signs of infestation?

Solution. Let A_i be the event that the ith tissue extracted from the ivy plant is infested by spider mites, $i = 1, 2, \ldots, 6$. The plant is infested in 20% of its area, so $\mathrm{P}(A_i) = 0.2$ and $\mathrm{P}(A_i') = 1 - 0.2 = 0.8$.

(a) Let B be the event that four successive samples show signs of infestation. By independence,

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 \cap A_4) = 0.2 \times 0.2 \times 0.2 \times 0.2 = 0.2^4 = 0.001024.$$

(b) For exactly 3 infested out of 4 successive samples, one sample is not infested. A single such arrangement (for instance $A_1 \cap A_2 \cap A_3 \cap A_4'$) has probability

$$0.2^3 \times 0.8 = 0.008 \times 0.8 \quad \Rightarrow \quad 0.2^3 \times 0.8^1.$$

We can select which 4 of the samples are taken and which is the uninfested one. Picking 3 out of 4 can be done in $\binom{4}{3} = 4$ ways, giving 4 disjoint events:

$$\mathrm{P} = \binom{4}{3} (0.2)^3 (0.8)^1 = 4 \times 0.008 \times 0.8 = 0.003072.$$

If we instead consider all 6 tissues with $n_1 \times n_2 = 6$ distinct arrangements but with 2 counted, the corresponding figure is 0.0768.

Game Defeat Problem

Example 7.18. When a player is defeated by an opponent, the game ends. The probability of a win is $\mathrm{P}(\text{win}) = 80\% = 0.80$, and $n = 4$. (a) What is the probability that a player defeats all four opponents in a game? (b) What is the probability of at least two wins in the last three sequences? (c) What is the probability of at least one occurrence of four wins?

Solution. The results from opponents are independent. The player defeats all 4 opponents if he wins each of 4 games.

(a)

$$\mathrm{P}(\text{defeat all 4}) = 0.8 \times 0.8 \times 0.8 \times 0.8 = 0.8^4 = 0.4096.$$

(b) The possible game sequences (with W = win, L = lose) are

$$L, \quad WL, \quad WWL, \quad WWWL, \quad WWWW.$$

With $\mathrm{P}(L) = 1 - \mathrm{P}(\text{win}) = 0.20$ and $\mathrm{P}(WL) = \mathrm{P}(\text{win}) \times \mathrm{P}(\text{lose}) = 0.8 \times 0.2 = 0.16$,

$$\mathrm{P}(\text{at least 2 wins}) = 1 - 0.2 - 0.16 = 0.64.$$

Equivalently, $\mathrm{P}(WWL + WWWL + WWWW) = 0.128 + 0.1024 + 0.4096 = 0.64$.

(c) The probability of not having four wins in a single attempt is

$$\mathrm{P}(\text{not 4 wins}) = 1 - \mathrm{P}(4 \text{ wins}) = 1 - 0.4096 = 0.5904.$$

Over three independent attempts,

$$\mathrm{P}(3 \text{ times not 4 wins}) = 0.5904 \times 0.5904 \times 0.5904 = 0.2058.$$

Therefore

$$\mathrm{P}(\text{at least once 4 wins}) = 1 - \mathrm{P}(3 \text{ times not 4 wins}) = 1 - 0.2058 = 0.7942.$$

Credit-Card Hacker

Example 7.19. A hacker attempts to guess a credit card number, where $N =$ one million valid numbers. (a) What is the probability of a correct guess on a single random selection? (b) Let B denote the event that the hacker correctly selects the month and year of credit card expiration. What is this probability?

Solution. **(a)**

$$\frac{1 \times 10^6}{10^{16}} = 10^{-10}.$$

(b) A hacker randomly selects one of 12 months, so the probability of correctly guessing the month is $\frac{1}{12}$. Combined with the year guess,

$$\mathrm{P}(B) = 0.25 \times \frac{1}{12} = \frac{1}{4} \times \frac{1}{12} = \frac{1}{48}.$$

Series–Parallel Circuit Operation

Example 7.20. The following circuit operates if and only if there is a path of functional devices from left to right. The probability that each device functions is as shown. Assume that the probability that a device functions does not depend on whether or not other devices are functional. What is the probability that the circuit operates?

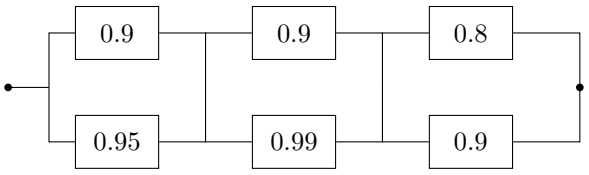


Figure 7.1: Three parallel blocks connected in series.

Solution. Each of the three blocks is a parallel pair; a block fails only if both of its devices fail. The block reliabilities are

$$1 - (1 - 0.9)(1 - 0.95) = 0.995, \quad 1 - (1 - 0.9)(1 - 0.99) = 0.999 \approx 0.995,$$

and the third block contributes $1 - (1 - 0.8)(1 - 0.9) = 0.98$. Since the blocks are in series,

$$\mathrm{P}(\text{operates}) = 0.995 \times 0.995 \times 0.98 = 0.9702.$$

7.4 Bayes' Theorem

If $E_1, E_2, \ldots, E_k$ are k mutually exclusive and exhaustive events and B is any event with $\mathrm{P}(B) > 0$, then

$$\mathrm{P}(E_i \mid B) = \frac{\mathrm{P}(B \mid E_i)\,\mathrm{P}(E_i)}{\mathrm{P}(B \mid E_1)\,\mathrm{P}(E_1) + \mathrm{P}(B \mid E_2)\,\mathrm{P}(E_2) + \cdots + \mathrm{P}(B \mid E_k)\,\mathrm{P}(E_k)}.$$

Preliminary Product Design Reviews

Example 7.21. Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor. (a) What is the probability that a product attains a good review? (b) If a new design attains a good review, what is the probability that it will be a highly successful product? (c) If a product does not attain a good review, what is the probability that it will be a highly successful product?

Solution. Let G be a good review with G' its complement, and let A_1, A_2, A_3 denote highly, moderately, and poorly successful products.

(a) By the total probability rule,

$$\mathrm{P}(G) = \sum_{j=1}^{3} \mathrm{P}(G \mid A_j)\,\mathrm{P}(A_j) = 0.95 \times 0.4 + 0.6 \times 0.35 + 0.1 \times 0.25 = 0.6150.$$

(b) By Bayes' theorem,

$$\mathrm{P}(A_1 \mid G) = \frac{\mathrm{P}(G \mid A_1)\,\mathrm{P}(A_1)}{\mathrm{P}(G)} = \frac{0.95 \times 0.4}{0.6150} = 0.6179.$$

(c)

$$\mathrm{P}(A_1 \mid G') = \frac{\mathrm{P}(G' \mid A_1)\,\mathrm{P}(A_1)}{\mathrm{P}(G')} = \frac{0.05 \times 0.4}{1 - 0.6150} = 0.052.$$

Inspector Identifying Defective Items

Example 7.22. An inspector working for a manufacturing company has a 99% chance of correctly identifying a defective item and a 0.5% chance of incorrectly classifying a good item as defective. The company has evidence that its line produces 0.9% of nonconforming items. (a) What is the probability that an item selected for inspection is classified as defective? (b) If an item is selected at random and classified as nondefective, what is the probability that it is good?

Solution. Let D be the event that a defective item is selected and let I be the event that an item is inspected and found defective, with D' being a nondefective item. Then

$$\mathrm{P}(I \mid D) = 0.99, \quad \mathrm{P}(I \mid D') = 0.005, \quad \mathrm{P}(D) = 0.009, \quad \mathrm{P}(D') = 0.991.$$

(a) By the total probability rule,

$$\mathrm{P}(I) = \mathrm{P}(I \mid D)\,\mathrm{P}(D) + \mathrm{P}(I \mid D')\,\mathrm{P}(D') = 0.99 \times 0.009 + 0.005 \times 0.991 = 0.0138.$$

(b) By Bayes' theorem,

$$\mathrm{P}(D' \mid I') = \frac{\mathrm{P}(I' \mid D')\,\mathrm{P}(D')}{\mathrm{P}(I')} = \frac{(1-0.005)(0.991)}{1-0.0138} = \frac{0.94 \times 0.99}{0.9309} \approx 0.999.$$

Disease Blood Test

Example 7.23. Suppose it is known that 1% of the population suffers from a particular disease. A blood test has a 97% chance of identifying the disease for diseased individuals, but also has a 6% chance of falsely indicating that a healthy person has the disease. (a) What is the probability that a person will have a positive blood test? (b) If your blood test is positive, what is the chance that you have the disease? (c) If your blood test is negative, what is the chance that you do not have the disease?

Solution. Let D be the event that a person has the disease and I the event that the test identifies the disease (tests positive), with

$$\mathrm{P}(D) = 0.01, \quad \mathrm{P}(D') = 0.99, \quad \mathrm{P}(I \mid D) = 0.97, \quad \mathrm{P}(I \mid D') = 0.06.$$

(a)

$$\mathrm{P}(I) = \mathrm{P}(I \mid D)\,\mathrm{P}(D) + \mathrm{P}(I \mid D')\,\mathrm{P}(D') = 0.97 \times 0.01 + 0.06 \times 0.99 = 0.0691.$$

(b) By Bayes' theorem,

$$\mathrm{P}(D \mid I) = \frac{\mathrm{P}(I \mid D)\,\mathrm{P}(D)}{\mathrm{P}(I)} = \frac{0.97 \times 0.01}{0.0691} = 0.1404.$$

(c)

$$\mathrm{P}(D' \mid I') = \frac{\mathrm{P}(I' \mid D')\,\mathrm{P}(D')}{\mathrm{P}(I')} = \frac{(1-0.06)(1-0.01)}{1-0.0691} = \frac{0.94 \times 0.99}{0.9309} = 0.9997.$$

Phone-Card Fraud Detection

Example 7.24. Software to detect fraud in consumer phone cards tracks the number of metropolitan areas where calls originate each day. It is found that 1% of the legitimate users originate calls from two or more metropolitan areas in a single day. However, 30% of fraudulent users originate calls from two or more metropolitan areas in a single day. The proportion of fraudulent users is 0.01%. If the same user originates calls from two or more metropolitan areas in a single day, what is the probability that the user is fraudulent?

Solution. Let S be the event that calls originate from two or more areas and let G be the event that the caller is legitimate, with

$$\mathrm{P}(S \mid G) = 0.01, \quad \mathrm{P}(S \mid G') = 0.3, \quad \mathrm{P}(G') = 0.0001.$$

By Bayes' theorem,

$$\mathrm{P}(G' \mid S) = \frac{\mathrm{P}(S \mid G')\,\mathrm{P}(G')}{\mathrm{P}(S \mid G')\,\mathrm{P}(G') + \mathrm{P}(S \mid G)\,\mathrm{P}(G)}$$

$$= \frac{0.3 \times 0.0001}{(0.3 \times 0.0001) + (0.01)(1-0.0001)} = \frac{0.00003}{0.00003 + 0.0099} = 0.2326.$$

7.5 Further Worked Examples

Machine Tool Idle Time

Example 7.25. A machine tool is idle 15% of the time. You request immediate use of the tool on 5 different occasions during the year. Assume that your requests represent independent events. (a) What is the probability that the tool is idle at the time of all of your requests? (b) What is the probability that the tool is idle at the time of exactly 4 of your requests? (c) What is the probability that the tool is idle at the time of at least 3 of your requests?

Solution. Let the probability that the tool is idle be 0.15.

(a) $(0.15)^5 = 7.6 \times 10^{-5}$,

(b) $\left[(0.15)^4 \times 0.85\right] \times 5 = 0.00215156$,

(c) $\mathrm{P}(\text{idle at least 3 times}) = \mathrm{P}(\text{exactly } 3) + \mathrm{P}(4) + \mathrm{P}(5)$
$= 10 \times (0.15)^3 \times (0.85)^2 + 5 \times (0.15)^4 \times 0.85 + (0.15)^5.$

Database Transactions

Example 7.26. Transactions to a computer database are either new items or changes to previous items. The addition of an item can be completed in less than 100 milliseconds 90% of the time, but only 20% of changes to a previous item can be completed in less than this time. If 30% of transactions are changes, what is the probability that a transaction can be completed in less than 100 milliseconds?

Solution. By the total probability rule, with 30% changes and 70% additions,

$$\mathrm{P}(< 100 \text{ ms}) = (0.2 \times 0.3) + (0.7 \times 0.9) = 0.06 + 0.63 = 0.69.$$

Automated Filling Operation

Example 7.27. In an automated filling operation, the probability of an incorrect fill when the process is operated at a low speed is 0.001. When the process is operated at high speed, the probability of an incorrect fill is 0.01. Assume that 30% of the containers are filled when the process is operated at a high speed and the remainder are filled when the process is operated at a low speed. (a) What is the probability of an incorrectly filled container? (b) If an incorrectly filled container is found, what is the probability that it was filled during high-speed operation?

Solution. Let D denote the event that a container is incorrectly filled and let H denote the event that a container is filled under high-speed operation. Then $\mathrm{P}(D \mid H) = 0.01$, $\mathrm{P}(D \mid H') = 0.001$, $\mathrm{P}(H) = 0.30$, and $\mathrm{P}(H') = 0.70$.

(a)

$$\mathrm{P}(D) = \mathrm{P}(D \mid H)\,\mathrm{P}(H) + \mathrm{P}(D \mid H')\,\mathrm{P}(H') = 0.01 \times 0.30 + 0.001 \times 0.70 = 0.0037.$$

(b) By Bayes' theorem,

$$\mathrm{P}(H \mid D) = \frac{\mathrm{P}(D \mid H)\,\mathrm{P}(H)}{\mathrm{P}(D)} = \frac{0.01 \times 0.30}{0.0037} = 0.8108.$$

Email Routing Through Two Server Routes

Example 7.28. An email message can travel through one of two server routes. The probability of transmission error in each of the servers and the proportion of messages that travel each route are shown in the following table. Assume that the servers are independent. (a) What is the probability that a message will arrive without error? (b) If a message arrives in error, what is the probability that it was sent through route 1?

	% of msgs	Probability of error (Server)			
		1	2	3	4
Route 1	30	0.01	0.015		
Route 2	70			0.02	0.003

Table 7.1: Routing probabilities and per-server error probabilities.

Solution. Let E represent an error in the message. A message arrives without error if every server on its route transmits without error. Therefore

$$\mathrm{P}(E') = (0.3)(0.99)(0.985) + (0.7)(0.98)(0.997) = 0.9764.$$

(b) By Bayes' theorem,

$$\mathrm{P}(\text{route } 1 \mid E) = \frac{\mathrm{P}(E \mid \text{route } 1)\,\mathrm{P}(\text{route } 1)}{\mathrm{P}(E)} = \frac{0.3 \times \big[1 - (0.99)(0.985)\big]}{1 - 0.9764}.$$

The numerator uses the route-1 selection probability of 0.3, and the denominator is $\mathrm{P}(E) = 1 - \mathrm{P}(E') = 1 - 0.9764$.